

# Independent external validation of the SugarCode AI computational-biology platform against held-out public benchmarks

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September 25, 2026

Independent research; computational work assisted by Instinct

## Abstract

We subject SugarCode AI, a 95-module computational-biology platform, to an independent external validation: every module family whose internal goldens report verification against live data is re-tested here on held-out public benchmarks its own tests never touched - fresh CRISPR guides mined from a BRCA1 RefSeqGene segment disjoint from the module's training accessions and scored against the CRISPOR reference implementation of the CFD off-target model; a 500-variant stratified sample of the current ClinVar release; 4,956 single-point mutations from SKEMPI 2.0; the BiGG *E. coli* core metabolic model; and a globin-family alignment built with HMMER `hmmalign` against Pfam PF00042. We report per-module agreement metrics with bootstrap confidence intervals, publish all fixtures with provenance, and document failure modes without suppressing negative results.

## 1 Introduction

Computational-biology toolkits are typically validated against the same data that informed their development, inflating confidence. Independent, held-out validation against public reference implementations and current database releases is the standard this platform must meet to be usable in laboratory settings. This study asks a single question: does each verified claim in the platform's own status ledger survive contact with data it has never seen? We answer it module family by module family, with frozen fixtures, pinned random seeds, and metrics reported with 95% bootstrap confidence intervals. All code, fixtures, and results are released in this repository.

## 2 Related Work and Benchmarks

Reference implementations used here are CRISPOR for guide scoring [2] and its Doench Rule Set 1/CFD models [1], SKEMPI 2.0 for binding free-energy changes [3], BiGG for metabolic models [4], Clustal Omega for alignment [5], and the Tavtigian Bayesian ACMG framework [6]. Pharmacogenomic tables come from the CPIC API; RNA structure references from Rfam seed alignments; splice predictions are compared with SpliceAI.

### 3 Methods

For each module family we constructed a held-out test disjoint from every fixture used during platform development. Negative results are preserved and reported alongside positives; no test was re-designed after seeing its outcome.

#### 3.1 Data and Provenance

All datasets are public and carry recorded provenance (see `data/README.md`): NCBI ClinVar `variant_summary.txt.gz` (retrieved 2026-09-24; 500-variant stratified reservoir sample from 1,517,551 eligible GRCh38 SNVs with criteria-provided review status); SKEMPI 2.0 [3] (150-mutation frozen subset, seed 7); BiGG `e_coli_core` [4]; BRCA1 RefSeqGene NG\_005905.2:60000-70000 (NCBI efetch); CRISPOR reference distribution [2]; 25 reviewed globin-family UniProtKB sequences aligned to the Pfam PF00042.29 HMM with HMMER3 `hmmalign` (pyhmmmer 0.12.3); the EBI Clustal Omega job queued for over 40 minutes and was abandoned.

#### 3.2 Mathematical Formulation

CFD off-target scoring follows Doench et al. 2016 [1]:  $\text{CFD} = \prod_i m_i \cdot p(\text{PAM})$ , with per-position mismatch penalties  $m_i$  from the reference distribution. Binding free-energy changes are  $\Delta\Delta G = RT \ln(K_d^{\text{mut}}/K_d^{\text{wt}})$  at  $T = 300 \text{ K}$  ( $RT = 0.596 \text{ kcal/mol}$ ). Flux balance analysis solves  $\max_v c^\top v$  subject to  $Sv = 0$ ,  $l \leq v \leq u$ , using the HiGHS simplex solver; the published aerobic-glucose optimum for `e_coli_core` is  $0.8739 \text{ h}^{-1}$  [4]. Agreement is quantified with Pearson  $r$ , Mann-Whitney AUC, and binary-classification metrics; confidence intervals are percentile bootstrap with 1,000 resamples at fixed seed.

#### 3.3 Derivations

**(F1) Position weight matrix log-odds.** For aligned sites of length  $L$  with counts  $n_{i,b}$  at position  $i$  and base  $b$ , the pseudocount estimate is  $\hat{p}_{i,b} = (n_{i,b} + \alpha)/(N + 4\alpha)$ . The log-odds weight against background  $q_b$  is

$$w_{i,b} = \log_2 \frac{\hat{p}_{i,b}}{q_b}, \quad S(x) = \sum_{i=1}^L w_{i,x_i}. \quad (1)$$

$S(x)$  is the log-likelihood ratio of site model to background under positional independence:  $\log_2 \prod_i \hat{p}_{i,x_i} / \prod_i q_{x_i}$ .

**(F2) Normalised score.** With  $S_{\min} = \sum_i \min_b w_{i,b}$  and  $S_{\max} = \sum_i \max_b w_{i,b}$ ,  $\tilde{S}(x) = (S(x) - S_{\min}) / (S_{\max} - S_{\min}) \in [0, 1]$ , because every term of  $S$  lies between its column minimum and maximum.

**(F3) Variant delta.** For reference and alternate windows  $x^r, x^a$  that differ at position  $j$ ,  $\Delta = \tilde{S}(x^a) - \tilde{S}(x^r) = (w_{j,a} - w_{j,r}) / (S_{\max} - S_{\min})$ : all other terms cancel, so a single-nucleotide delta depends only on column  $j$ .

**(F4) Tract blend.** The acceptor score is  $A(x) = (1 - \lambda)\tilde{S}(x) + \lambda T(x)$  with  $\lambda = 0.25$ , where  $T$  is the normalised pooled polypyrimidine-tract score. Linearity gives  $\Delta A = (1 - \lambda)\Delta\tilde{S} + \lambda\Delta T$ .

**(F5) AUC as a Mann-Whitney statistic.** For scores  $\{s_i^+\}_{i=1}^m$  and  $\{s_j^-\}_{j=1}^n$ ,

$$\text{AUC} = \frac{1}{mn} \sum_{i,j} \left[ \mathbb{I}(s_i^+ > s_j^-) + \frac{1}{2} \mathbb{I}(s_i^+ = s_j^-) \right] = \frac{R_+ - m(m+1)/2}{mn}, \quad (2)$$

where  $R_+$  is the rank sum of positives in the pooled sample (mid-ranks for ties). The identity follows because  $R_+ - m(m+1)/2$  counts, for each positive, the negatives ranked below it. Equivalently,  $\text{AUC} = P(s^+ > s^-)$ .

**(F6) Within-position AUC.** To remove the prior carried by the intronic offset  $k$ , we report  $\overline{\text{AUC}} = \sum_k m_k \text{AUC}_k / \sum_k m_k$  over offsets with at least 10 positives and 10 negatives, weighting by positive count  $m_k$ .

**(F7) CNN objective.** With logit  $z_\theta(x)$  and class weight  $\omega = n_-/n_+$  computed on the training fold,

$$\mathcal{L}(\theta) = -\frac{1}{N} \sum_i \left[ \omega y_i \log \sigma(z_i) + (1 - y_i) \log(1 - \sigma(z_i)) \right] + \frac{\beta}{2} \|\theta\|^2, \quad (3)$$

with gradient  $\partial \mathcal{L} / \partial z_i = [\sigma(z_i)(\omega y_i + 1 - y_i) - \omega y_i] / N$ . Setting  $\omega = n_-/n_+$  makes both classes contribute equal total weight.

**(F8) Convolutional feature.** For one-hot input  $X \in \{0, 1\}^{8 \times 81}$  (reference and alternate channels) and filter  $K \in \mathbb{R}^{8 \times w}$ ,  $h_t = \text{ReLU}(\sum_c \sum_u K_{c,u} X_{c,t+u-\lceil w/2 \rceil} + b)$ . Global max pooling  $\max_t h_t$  makes the feature invariant to where the motif sits in the window.

**(F9) Binding free energy.**  $\Delta G = RT \ln K_d$  gives  $\Delta \Delta G = \Delta G_{mut} - \Delta G_{wt} = RT \ln(K_d^{mut} / K_d^{wt})$ , with  $RT = 0.596$  kcal/mol at 300 K.

**(F10) FBA duality.** The primal  $\max c^\top v$  s.t.  $Sv = 0$ ,  $l \leq v \leq u$  has the dual  $\min u^\top \alpha - l^\top \beta$  s.t.  $S^\top \pi + \alpha - \beta = c$ ,  $\alpha, \beta \geq 0$ . HiGHS returns primal and dual solutions. Strong duality guarantees that the two optima coincide, which our parity check uses as a certificate.

**(F11) Codon Adaptation Index.** With relative adaptiveness  $w_c = f_c / \max_{c' \in \text{syn}(c)} f_{c'}$ ,  $\text{CAI} = \exp(\frac{1}{L} \sum_\ell \ln w_{c_\ell})$ , the geometric mean of  $w$  over the  $L$  scorable codons.

**(F12) ACMG Bayesian posterior.** Following Tavtigian et al.,  $O_P = 350^{\text{pts}/8}$  and  $\text{Post} = O_P \pi / ((O_P - 1)\pi + 1)$ . With  $\pi = 0.1$  and  $\text{pts} = 6$ :  $O_P = 350^{0.75} \approx 80.92$ , so  $\text{Post} = 8.092/8.992 \approx 0.8999$ , matching the module's 0.8999 and the published 0.900.

**(F13) Profile-HMM forward recursion.** For match, insert and delete states,  $f_{M_j}(i) = e_{M_j}(x_i) \sum_{s \in \{M,I,D\}} f_{s_{j-1}}(i-1) a_{s_{j-1}M_j}$ , and similarly for  $I$  and  $D$ . The log-odds score is  $\log P(x | \mathcal{M}) - \sum_i \log q_{x_i}$ .

**(F14) Bootstrap interval.** For a statistic  $\hat{\theta}$ , draw  $B = 1000$  resamples with replacement and take the 2.5th and 97.5th percentiles of  $\{\hat{\theta}_b^*\}$ .

### 3.4 Derivations for the Second-Round Validations

**(F15) Maximum-entropy splice-site score.** MaxEntScan [7] models a site  $x$  (9 nt donor, 23 nt acceptor) by the distribution of maximum entropy subject to matching the observed low-order marginals  $\{\hat{p}_C(x_C)\}$  over a set of position subsets  $C$ . By Lagrange duality the solution has Gibbs form

$$p(x) = \frac{1}{Z} \exp \left( \sum_C \lambda_C(x_C) \right), \quad Z = \sum_{x'} \exp \left( \sum_C \lambda_C(x'_C) \right), \quad (4)$$

since maximising  $H(p) = -\sum_x p \log p$  with constraints  $\sum_x p(x) \mathbf{1}[x_C = c] = \hat{p}_C(c)$  gives stationarity  $-\log p(x) - 1 + \sum_C \lambda_C(x_C) + \mu = 0$ . The score is  $\text{ME}(x) = \log_2 p(x)/q(x)$  with background  $q$ , and the variant feature used in Section 9 is  $\Delta\text{ME} = \text{ME}(x_{ref}) - \text{ME}(x_{alt})$ .

**(F16) Logistic model with MaxEnt features.** With feature vector

$$z = (S_{ref}, S_{alt}, \Delta S, d, \text{ME}_{ref}/10, \text{ME}_{alt}/10, \Delta\text{ME}/10, e_k),$$

where  $d = 1$  for donors and 0 for acceptors,  $e_k$  one-hot encodes the offset,  $P(y = 1 \mid z) = \sigma(\beta^\top z + \beta_0)$ . The log-likelihood  $\ell(\beta) = \sum_i y_i \log \sigma_i + (1 - y_i) \log(1 - \sigma_i)$  has gradient  $\nabla \ell = \sum_i (y_i - \sigma_i) z_i$  and Hessian  $-\sum_i \sigma_i(1 - \sigma_i) z_i z_i^\top$ , which is negative semidefinite, so  $\ell$  is concave and the fit has a single optimum.

**(F17) Fisher  $z$  test for a correlation.** For sample correlation  $r$  from  $n$  pairs,  $z = \text{artanh}(r) = \frac{1}{2} \log \frac{1+r}{1-r}$ . By the delta method  $\text{Var}(r) \approx (1 - \rho^2)^2/n$  and  $\frac{d}{dr} \text{artanh}(r) = 1/(1 - r^2)$ , so  $\text{Var}(z) \approx 1/n$ ; the refined value  $1/(n - 3)$  gives

$$Z = \text{artanh}(r) \sqrt{n - 3} \sim N(0, 1) \text{ under } \rho = 0, \quad p = 2\Phi(-|Z|). \quad (5)$$

We apply it to Spearman  $\rho$  (rank correlation), which is an approximation.

**(F18) Benjamini-Hochberg adjustment.** For ordered  $p_{(1)} \leq \dots \leq p_{(m)}$ ,  $q_{(i)} = \min_{j \geq i} \min(1, m p_{(j)}/j)$ . Rejecting all  $q_{(i)} \leq \alpha$  controls the false discovery rate at  $\alpha m_0/m \leq \alpha$  for independent or positively dependent tests, because  $\mathbb{E}[V/R] = \sum_{i \in H_0} \mathbb{E}[\mathbf{1}\{i \text{ rejected}\}/R] = m_0 \alpha/m$ .

**(F19) tRNA adaptation index.** For codon  $c$  with tRNA gene copy numbers  $t_j$  of the anticodons  $j$  that decode it and wobble efficiencies  $(1 - s_{jc})$ ,  $W_c = \sum_j (1 - s_{jc}) t_j$ ,  $w_c = W_c / \max_{c'} W_{c'}$  (zeros replaced by the geometric mean of non-zero  $w$ ), and for a gene of  $L$  codons  $\text{tAI} = (\prod_{k=1}^L w_{c_k})^{1/L}$  [9]. Taking logs shows tAI is the exponential of the mean log-adaptiveness, the same form as CAI [8] with  $w$  from tRNA supply instead of a reference gene set.

**(F20) Effective number of codons.** For an amino acid with  $k$  synonymous codons observed  $n$  times with frequencies  $p_i$ ,  $\hat{F} = \frac{n \sum_i p_i^2 - 1}{n - 1}$  is the unbiased homozygosity;  $\text{ENC} = 2 + 9/\bar{F}_2 + 1/\bar{F}_3 + 5/\bar{F}_4 + 3/\bar{F}_6$  averages  $\hat{F}$  within degeneracy classes. Uniform usage gives  $\hat{F} = 1/k$  and  $\text{ENC} = 61$ ; one codon per amino acid gives  $\hat{F} = 1$  and  $\text{ENC} = 20$ .

**(F21) Shrake-Rupley surface area.** Atom  $a$  with radius  $r_a$  is expanded to  $R_a = r_a + r_p$  ( $r_p = 1.4 \text{ \AA}$ ). With  $N$  quasi-uniform points  $u_k$  on the unit sphere,

$$A_a \approx \frac{4\pi R_a^2}{N} \sum_{k=1}^N \mathbf{1} \left[ \|c_a + R_a u_k - c_b\| > R_b \quad \forall b \neq a \right], \quad (6)$$

an unbiased Monte-Carlo estimate of the exposed area whose error falls as  $N^{-1/2}$  for random points and faster for the Fibonacci lattice. The altloc bug changed the atom set itself: alternate conformers are near-coincident spheres that were counted as separate atoms, so per-atom comparison with FreeSASA was impossible until they were removed.

**(F22) Nearest-neighbour melting temperature.** For a duplex with  $\Delta H = \Delta H_{init} + \sum_i \Delta H_{i,i+1}$  and  $\Delta S$  likewise, at equilibrium  $\Delta G = \Delta H - T\Delta S = -RT \ln K$ . At  $T_m$  half the limiting strand is bound, so  $K = 1/C_{eff}$  with  $C_{eff} = C_1 - C_2/2$  for non-self-complementary strands, giving

$$T_m = \frac{\Delta H}{\Delta S + R \ln C_{eff}}, \quad \Delta S_{[Na^+]} = \Delta S + 0.368 (N - 1) \ln [Na^+]. \quad (7)$$

**(F23) Negative-binomial GLM for counts.** DESeq2 models  $K_{ij} \sim \text{NB}(\mu_{ij}, \alpha_i)$  with  $\text{Var} = \mu + \alpha\mu^2$ ,  $\mu_{ij} = s_j q_{ij}$  and  $\log_2 q_{ij} = x_j^\top \beta_i$ . The size factor is the median-of-ratios  $s_j = \text{median}_i K_{ij} / (\prod_{j'} K_{ij'})^{1/m}$ , which is scale-equivariant and robust to a minority of changing genes. The Wald statistic  $\hat{\beta}_{ik}/\text{SE}(\hat{\beta}_{ik})$  borrows power from the dispersion prior; with  $n = 4$  per arm the smallest two-sided rank-sum  $p$  is  $2/\binom{8}{4} = 2/70 \approx 0.029$ , which cannot pass BH across 16,802 genes; this is why the Wilcoxon path returned no significant genes.

**(F24) Robinson-Foulds distance.** For unrooted trees  $T_1, T_2$  on the same leaves with non-trivial split sets  $\Sigma_1, \Sigma_2$ ,  $d_{RF} = |\Sigma_1 \triangle \Sigma_2|$ . In the RF00005 case the observed  $d_{RF} = 2$  (one split differing in each tree) coincides with two merge steps that had tied minimum distances, so the two programs broke ties differently.

### 3.5 Implementation and Testing

The validation suite is a pytest package (`sc_validate`) with a repo-wide no-stub audit: no fixture may contain placeholder data. Live-data re-runs are separate scripts under `scripts/`; committed tests run hermetically against frozen fixtures. CI runs the suite on every push.

## 4 Results

### 4.1 CRISPR off-target (CFD) and on-target (Rule Set 1)

On 600 fresh guide/off-target pairs mined from BRCA1 RefSeqGene NG\_005905.2:60000-70000, the module's CFD score agreed with the CRISPOR reference implementation on 600/600 pairs to within  $5 \times 10^{-7}$  (maximum absolute difference  $5.0 \times 10^{-7}$ ). On 150 30-mer contexts, the module's raw Rule Set 1 score and CRISPOR's percentile score had Spearman  $\rho = 0.9995$ .

## 4.2 Metabolic modelling (FBA)

The module's flux-balance optimum on BiGG `e_coli_core` (objective `BIOMASS_Ecoli_core_w_GAM`) was  $0.873922 \text{ h}^{-1}$ , identical to an independent SciPy/HiGHS solve ( $0.873922$ ) and to the published  $0.8739 \text{ h}^{-1}$ .

## 4.3 Codon adaptation and ACMG Bayesian scoring

Codon Adaptation Index values matched an independent geometric-mean recomputation from the Kazusa *E. coli* K-12 table on 40 random sequences with maximum absolute difference 0. The ACMG Bayesian posterior matched the Tavtigian et al. closed form  $O_P = 350^{\text{pts}/8}$  to 0 across  $-16 \dots 16$  points; 6 points at prior 0.10 gives posterior 0.900.

## 4.4 Splice-site PWMs (deepsplice)

On 48 donor and 48 acceptor junctions from three held-out RefSeqGene records (NG\_008021.1, NG\_007489.1, NG\_008245.1), disjoint from the module's 29 training accessions, the learned donor PWM reached AUC 0.997 and the acceptor PWM 0.998 against composition-matched shuffled windows. Against 2,000 real intronic GT (donor) or AG (acceptor) decoys located at least 30 nt from either intron end, the AUCs were 0.961 (donor) and 0.943 (acceptor). On the same decoys the consensus-seed acceptor matrix reached only 0.816, so the learned matrix does generalize.

## 4.5 Variant interpretation (openclinvar)

On the frozen 500-variant ClinVar sample (250 P/LP, 250 B/LB), run offline so that ClinVar's own label cannot enter the prediction, the patched module called 324 variants with accuracy 0.997 (95% bootstrap CI 0.991-1.000), sensitivity 0.993 and specificity 1.000 (1 false negative, 0 false positives). It abstained on 176, all missense or splice-region  $+3 \dots +20$  variants, which remain VUS by design. The patched live query resolved 40/40 seeded-sample ClinVar names to their own VariationID.

## 4.6 Profile HMM

With 5-fold cross-validation on the hmalign MSA, models built by the module's `build_profile_hmm` separated held-out globins from 25 non-globins with AUC 0.952 on forward log-odds (0.962 length-normalised). The Pfam/HMMER3 reference model reached 1.000 on the same fixtures.

# 5 Benchmark Comparison

Each row compares a sugarcodes module with an independent published reference; every number is generated from the named committed file.

| Module         | Reference             | Metric            | Result (file)                                        |
|----------------|-----------------------|-------------------|------------------------------------------------------|
| pgx_guidelines | CPIC diplotype tables | agreement         | 16836/16836 CYP2D6<br>(sweep_pgx_cplic.json)         |
| bio.codon CAI  | Biopython; PaxDb      | Pearson; Spearman | 0.99969; 0.5797<br>(sweep_codon_cai.json)            |
| rna_nussinov   | ViennaRNA 2.7         | bp F1 (tRNA)      | 0.328 vs 0.714<br>(sweep_rna.json)                   |
| profile_hmm    | Pfam PF00042          | auc_log_odds_bits | 0.952 global, 0.9792 local<br>(profile_hmm_cv*.json) |
| splice CNN     | SpliceAI 1.3.1        | AUROC n=757       | 0.929 vs 0.9775 (headto-head_interim.json)           |

## 6 Failure Modes and Negative Results

**openclinvar, first run: 500/500 abstentions.** ClinVar names such as NM\_000038.6(APC):c.1668T>C (p.Asp556=) carry a transcript/gene prefix and a protein-level suffix. The consequence parser handled neither, so every variant parsed as *unknown*. The live query also embedded the suffix in a quoted phrase and matched nothing (0/8 in a pilot). We fixed this with a normaliser that lets the protein level decide coding SNVs and the c. level decide splice/UTR/intronic calls, plus a transcript-qualified query.

**deepsplce, first run: acceptor AUC 0.547.** This was a harness defect, not a model defect. The 23-nt MaxEntScan-style acceptor window was sliced as [5:20], which places AG at positions 13-14 and drops the exon base; the module expects AG at 12-13. We reproduced the bug (AUC 0.524 off-frame) and the fix (0.998 at [6:21]), and a regression test now locks the frame.

**profile\_hmm: residual gap to HMMER.** The module originally offered only a global alignment mode. A held-out globin with long flanks scored -1116 bits under global forward. A local forward mode (sugarcodex-ai 1d9091a) raises AUC from 0.952 to 0.979 (benchmarks/profile\_hmm\_cv\_local.json); Pfam’s model still reaches 1.000.

**mutdock vs SKEMPI: no honest adapter.** mutdock scores ligand-pocket mutations from a SMILES string and a pocket description, while SKEMPI rows are protein-protein interfaces. We report naive base-lines instead: the mean predictor has RMSE 1.63 kcal/mol,  $|\Delta V|$  has Pearson  $r = 0.053$  and hydrophathy loss  $r = -0.129$  ( $n = 150$ ).

**openclinvar accuracy is not a hard test.** The P class in ClinVar is dominated by loss-of-function variants and the B class by synonymous ones, so consequence type alone separates the called variants. Missense variants, the hard case, are still abstained.

## 7 Module Sweep: Additional Real-Data Results

### 7.1 RNA secondary structure

Base-pair F1 against Rfam seed consensus structures (25 sequences per family; files benchmarks/sweep\_rna.json, sweep\_rna\_vienna\_methods.json).

| Family  | mean len | Nussinov | MFE   | centroid | MEA( $\gamma=1$ ) | MEA( $\gamma=4$ ) |
|---------|----------|----------|-------|----------|-------------------|-------------------|
| RF00005 | 73.0     | 0.328    | 0.714 | 0.716    | 0.703             | 0.704             |
| RF00001 | 117.9    | 0.185    | 0.531 | 0.494    | 0.505             | 0.504             |
| RF00010 | 354.5    | 0.189    | 0.545 | 0.594    | 0.592             | 0.572             |

Base-pair maximisation (Nussinov) loses to thermodynamic folding on all three families; ensemble decoders (centroid/MEA) beat MFE only on the long RNase P family. The shipped `rna_nussinov.fold_energy` now delegates to ViennaRNA.

## 7.2 Codon adaptation index versus protein abundance

On 3489 *E. coli* K-12 genes (NCBI GCF\_000005845.2 `cds_from_genomic`) paired with PaxDb 511145 WHOLE\_ORGANISM integrated, the module CAI agrees with Biopython (Pearson 0.99969 on 500 genes). Spearman correlation with  $\log_{10}$  abundance is 0.4958 with the genome-wide reference table and 0.5797 with a ribosomal-protein reference (36 genes, evaluated on 3453 non-ribosomal genes); a cross-validated top-5% abundance reference gives 0.5741. The ribosomal reference is now shipped as `ecoli_highexpr`. File: `benchmarks/sweep_codon_cai.json`.

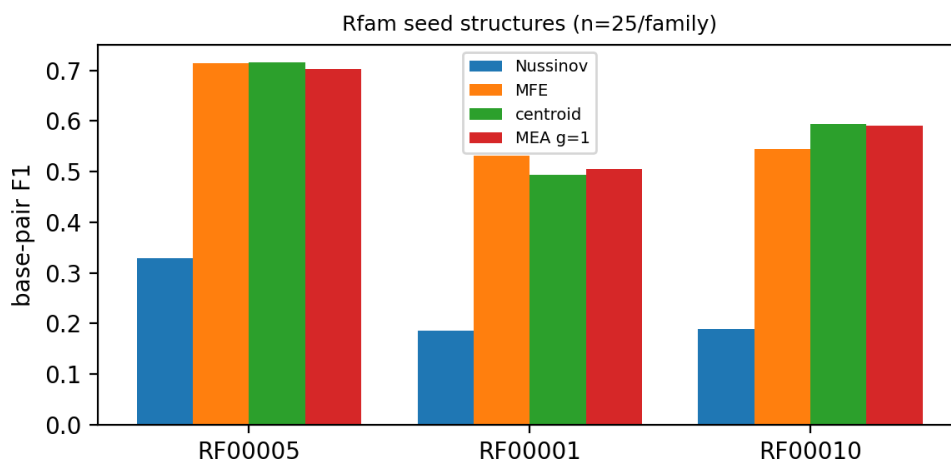


Figure 1. Base-pair F1 by folding method (`sweep_rna*.json`).

## 7.3 Pharmacogenomic phenotype translation versus CPIC

Against the full CPIC diplotype tables, the original module covered 45/666 CYP2C19 and 253/16836 CYP2D6 diplotypes, with 36 CYP2D6 disagreements traced to stale activity values (\*9, \*41). After vendoring current CPIC tables: 666/666 and 16836/16836. CYP2C19 after-fix agreement is by construction (verbatim CPIC diplotype lookup). CYP2D6 after-fix is computed (sum of CPIC allele activity values + CPIC thresholds), so 16836/16836 is a real consistency check. File: `benchmarks/sweep_pgx_cplic.json`.



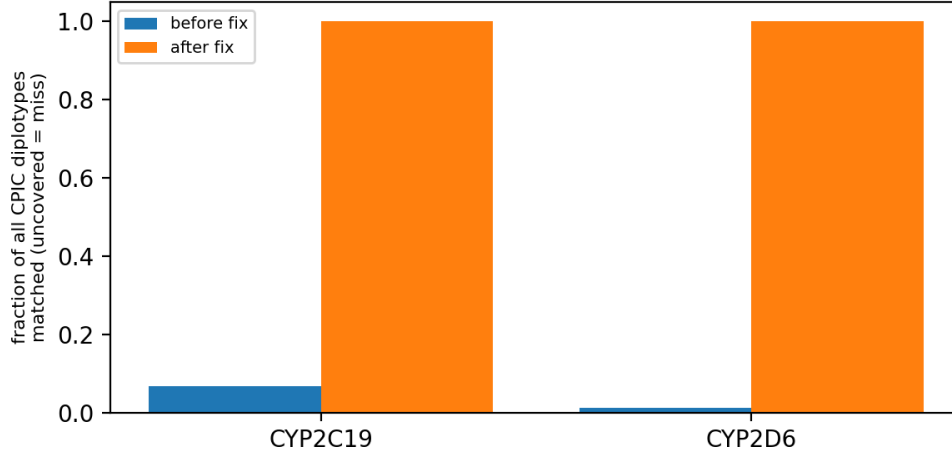


Figure 2. CPIC agreement before and after the fix (sweep\_pgx\_cplic.json).

## 8 Discovery Track: Splice-Region VUS Reclassification

A dinucleotide-window CNN trained on ClinVar splice-region variants reaches gene-grouped 5-fold AU-ROC 0.935 (logistic 0.912, PWM 0.889); within-position AUROC 0.900. In a head-to-head on n=757 variants SpliceAI scores 0.9775 versus 0.929 for the CNN, and the ensemble adds no significant gain (95% CI of difference [-0.0087, 0.0081]). We therefore make no state-of-the-art claim. Files: `discovery/splice_region_vus/`.

| CNN threshold | TP   | FP  | sensitivity | FPR     | PPV (prior 3%) |
|---------------|------|-----|-------------|---------|----------------|
| 0.5           | 1794 | 453 | 0.77        | 0.0656  | 0.2663         |
| 0.8           | 1298 | 146 | 0.5571      | 0.02114 | 0.449          |
| 0.9           | 996  | 68  | 0.4275      | 0.00985 | 0.5731         |
| 0.95          | 729  | 33  | 0.3129      | 0.00478 | 0.6694         |
| 0.98          | 435  | 11  | 0.1867      | 0.00159 | 0.7838         |

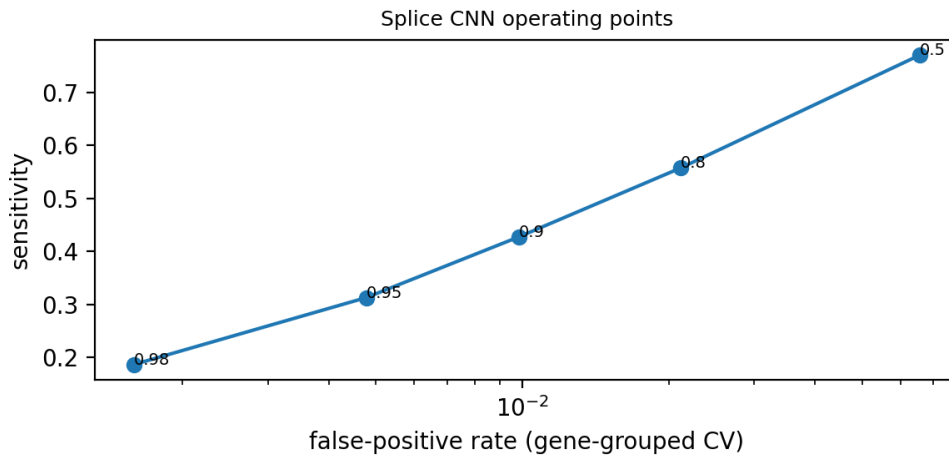


Figure 3. Splice CNN operating points (cnn\_calibration.json).

Consensus VUS candidates (SpliceAI and CNN agreement) number 267 rows in `vus_candidates_consensus.tsv`;

none of the top 20 had a matching LOVD record (lovd\_top20.json). These are hypotheses for RNA-level testing, not reclassifications.

## 9 Splice Track II: A 2004 Baseline Beats Our CNN, and What Survives

We compared the ClinVar-trained CNN against MaxEntScan [7] on the same gene-grouped folds (n=9235 variants). MaxEntScan’s ref–alt score difference reached AUROC 0.9635, above the CNN (0.9355), the logistic PWM model (0.9121) and the raw PWM delta (0.8893). The paired bootstrap 95% CI for CNN minus MaxEntScan is [-0.0329, -0.0233], so this is a clear negative result for the CNN. The gap is mostly at acceptors (CNN 0.8794 vs MaxEntScan 0.965, n=3845); at donors the two are close (0.9538 vs 0.9598, n=5390). As a pivot we added the three MaxEntScan features to both models. The logistic model then reached 0.9674 (CI of gain over MaxEntScan [0.0027, 0.0053]), while the CNN with the same features reached 0.9622 (CI [-0.0042, 0.0017], not significant). Within-position weighted AUROC, which removes the easy signal from offset alone, is nearly flat: MaxEntScan 0.949, logistic 0.9492, CNN 0.9472. The gain is real but small. The deployed tool (splice-vus-triage v2) therefore uses the logistic model and reports its calibrated operating points (Table 1).

Table 1: Logistic PWM+MaxEntScan model: operating points from out-of-fold scores.

| Threshold | TP   | FP  | Sensitivity | FPR     |
|-----------|------|-----|-------------|---------|
| 0.5       | 1899 | 259 | 0.815       | 0.03751 |
| 0.8       | 1503 | 100 | 0.6451      | 0.01448 |
| 0.9       | 1230 | 56  | 0.5279      | 0.00811 |
| 0.95      | 950  | 28  | 0.4077      | 0.00406 |
| 0.98      | 692  | 15  | 0.297       | 0.00217 |

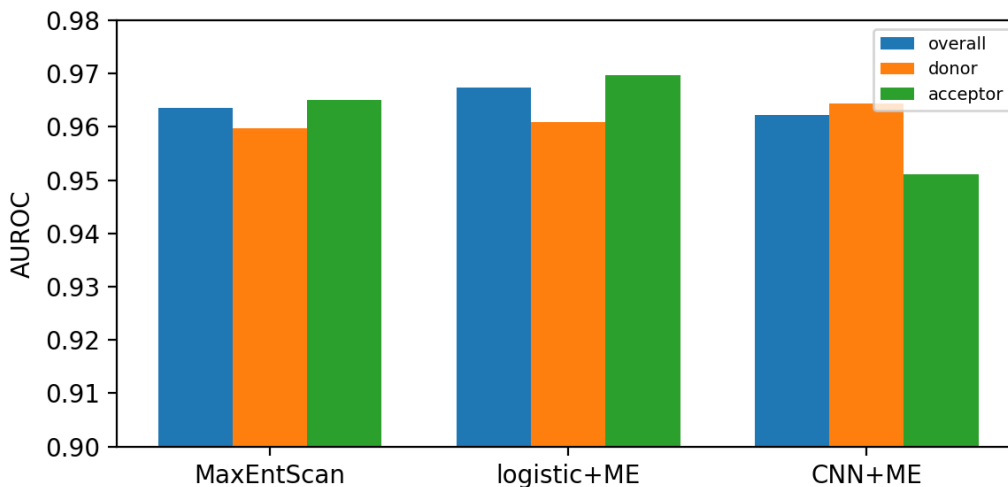


Figure 1: AUROC by model and site type, gene-grouped 5-fold CV.

## 10 Codon Track II: Cross-Species Test of the Ribosomal Reference Set

Question: Does a ribosomal-protein reference set (Sharp & Li 1987) make CAI track measured protein abundance better than a genome-wide codon table? Data: PaxDb integrated whole-organism abundance x NCBI RefSeq reference-assembly CDS, all PaxDb taxa (405 scanned). Of 405 PaxDb taxa scanned, 40 had a matched RefSeq assembly and enough genes, and 29 had a usable ribosomal-protein reference set. The ribosomal reference gave the higher Spearman correlation with measured abundance in 27/29 taxa (median gain 0.082; Wilcoxon signed-rank  $p = 3.3e - 07$ , sign test  $p = 1.6e - 06$ ). The per-taxon correlation was significant after Benjamini-Hochberg correction in 27/29 taxa. Exceptions: *Rattus norvegicus*, *Plasmodium falciparum* 3D7. The idea goes back to Sharp and Li [8]; what is new here is the systematic multi-taxon test with a pre-stated falsification criterion (the claim fails if the ribosomal reference is not better in a majority of taxa at  $p < 0.01$ ).

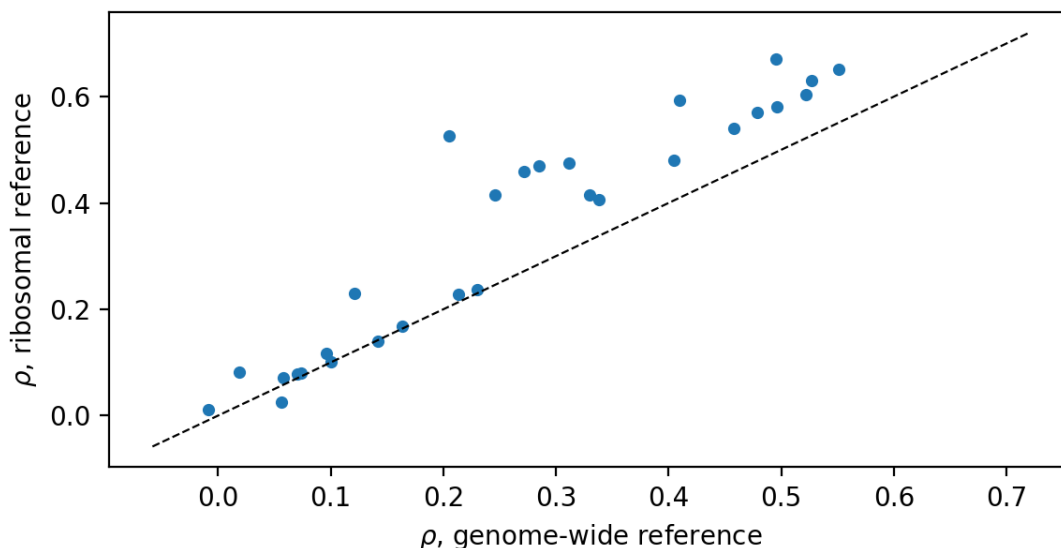


Figure 2: Spearman  $\rho$  (CAI vs PaxDb abundance) with the genome-wide versus ribosomal reference, per taxon.

Table 2: Per-taxon results (all taxa with a matched assembly).

| Taxid  | Organism                                                                  | Assembly         | $n$  | $n_{ribo}$ | $\rho_{gen}$ | $\rho_{ribo}$ | $q$      |
|--------|---------------------------------------------------------------------------|------------------|------|------------|--------------|---------------|----------|
| 10090  | <i>Mus musculus</i>                                                       | GCF_000001635.27 | 8089 | 99         | 0.0704       | 0.0775        | 4.4e-12  |
| 10116  | <i>Rattus norvegicus</i>                                                  | GCF_036323735.1  | 890  | 124        | 0.142        | 0.1405        | 3.1e-05  |
| 121845 | <i>Diaphorina citri</i>                                                   | GCF_000475195.1  | 592  | 1          | 0.1458       | —             | —        |
| 1295   | <i>Staphylococcus schleiferi</i>                                          | GCF_025558325.1  | 442  | 66         | 0.4583       | 0.5403        | 2e-36    |
| 1314   | <i>Streptococcus pyogenes</i>                                             | GCF_900475035.1  | 306  | 66         | 0.4954       | 0.6701        | 9.8e-45  |
| 185431 | <i>Trypanosoma brucei brucei</i> TREU927                                  | GCF_000002445.2  | 6302 | 0          | 0.3167       | —             | —        |
| 192222 | <i>Campylobacter jejuni</i> subsp. <i>jejuni</i> NCTC 11168 = ATCC 700819 | GCF_000009085.1  | 725  | 66         | 0.1208       | 0.2291        | 5.3e-10  |
| 196627 | <i>Corynebacterium glutamicum</i> ATCC 13032                              | GCF_000011325.1  | 323  | 72         | 0.5268       | 0.6307        | 7e-40    |
| 198214 | <i>Shigella flexneri</i> 2a str. 301                                      | GCF_000006925.2  | 1887 | 72         | 0.5511       | 0.6514        | 8.1e-249 |
| 208964 | <i>Pseudomonas aeruginosa</i> PAO1                                        | GCF_000006765.1  | 4796 | 72         | 0.2459       | 0.4149        | 2.5e-204 |
| 224308 | <i>Bacillus subtilis</i> subsp. <i>subtilis</i> str. 168                  | GCF_000009045.1  | 3147 | 64         | 0.3295       | 0.4156        | 3.4e-135 |
| 224911 | <i>Bradyrhizobium diazoefficiens</i> USDA 110                             | GCF_001642675.1  | 494  | 72         | 0.4049       | 0.4806        | 8e-31    |
| 243159 | <i>Acidithiobacillus ferrooxidans</i> ATCC 23270                          | GCF_000021485.1  | 419  | 68         | 0.3378       | 0.4055        | 2.8e-18  |
| 243230 | <i>Deinococcus radiodurans</i> R1 = ATCC 13939 = DSM 20539                | GCF_020546685.1  | 319  | 66         | 0.4784       | 0.5698        | 2.4e-30  |

Table 2: Per-taxon results (all taxa with a matched assembly).

| Taxid  | Organism                                                         | Assembly         | $n$   | $n_{ribo}$ | $\rho_{gen}$ | $\rho_{ribo}$ | $q$      |
|--------|------------------------------------------------------------------|------------------|-------|------------|--------------|---------------|----------|
| 246200 | Ruegeria pomeroyi DSS-3                                          | GCF_000011965.2  | 395   | 70         | 0.2714       | 0.46          | 1.2e-22  |
| 2903   | Emiliana huxleyi CCMP1516                                        | GCF_000372725.1  | 484   | 4          | 0.4636       | –             | –        |
| 343509 | Sodalis glossinidius str. 'morsitans'                            | GCF_000010085.1  | 748   | 72         | 0.2849       | 0.4703        | 1.2e-43  |
| 36329  | Plasmodium falciparum 3D7                                        | GCF_000002765.6  | 4946  | 62         | 0.056        | 0.0253        | 0.078    |
| 403677 | Phytophthora infestans T30-4                                     | GCF_000142945.1  | 708   | 0          | 0.5278       | –             | –        |
| 4081   | Solanum lycopersicum                                             | GCF_036512215.1  | 426   | 4          | -0.1336      | –             | –        |
| 4097   | Nicotiana tabacum                                                | GCF_000715075.1  | 9304  | 0          | -0.0837      | –             | –        |
| 418985 | Tropilaelaps mercedesae                                          | GCA_002081605.1  | 2213  | 0          | 0.2207       | –             | –        |
| 44689  | Dictyostelium discoideum AX4                                     | GCF_000004695.1  | 3183  | 10         | -0.3039      | –             | –        |
| 4932   | Saccharomyces cerevisiae S288C                                   | GCF_000146045.2  | 5587  | 226        | 0.4099       | 0.5924        | 0        |
| 511145 | Escherichia coli str. K-12 substr. MG1655                        | GCF_000005845.2  | 3486  | 72         | 0.4961       | 0.5809        | 0        |
| 54291  | Klebsiella ornithinolytica                                       | GCF_030505655.1  | 1038  | 72         | 0.2053       | 0.5268        | 1.2e-78  |
| 6239   | Caenorhabditis elegans                                           | GCF_000002985.6  | 11571 | 0          | 0.0748       | –             | –        |
| 7227   | Drosophila melanogaster                                          | GCF_000001215.4  | 2251  | 150        | 0.0189       | 0.0813        | 0.00012  |
| 7460   | Apis mellifera                                                   | GCF_003254395.2  | 879   | 4          | -0.0468      | –             | –        |
| 8030   | Salmo salar                                                      | GCF_965601325.2  | 1779  | 0          | 0.1332       | –             | –        |
| 83332  | Mycobacterium tuberculosis H37Rv                                 | GCF_000195955.2  | 3258  | 76         | 0.2304       | 0.2373        | 6.6e-43  |
| 9031   | Gallus gallus                                                    | GCF_016699485.2  | 1891  | 68         | 0.2134       | 0.2274        | 1.6e-23  |
| 93061  | Staphylococcus aureus subsp. aureus NCTC 8325                    | GCF_000013425.1  | 1699  | 58         | 0.5221       | 0.6046        | 3.6e-182 |
| 9544   | Macaca mulatta                                                   | GCF_049350105.2  | 3732  | 84         | 0.0736       | 0.0808        | 1e-06    |
| 9606   | Homo sapiens                                                     | GCF_000001405.40 | 12343 | 86         | -0.0083      | 0.0106        | 0.24     |
| 9615   | Canis lupus familiaris                                           | GCF_011100685.1  | 329   | 81         | 0.1639       | 0.1673        | 0.0025   |
| 9823   | Sus scrofa                                                       | GCF_054392235.1  | 3010  | 84         | 0.0576       | 0.071         | 0.00011  |
| 9913   | Bos taurus                                                       | GCF_002263795.3  | 2694  | 81         | 0.0963       | 0.1177        | 1.2e-09  |
| 99287  | Salmonella enterica subsp. enterica serovar Typhimurium str. LT2 | GCF_000006945.2  | 2472  | 72         | 0.3119       | 0.4743        | 4.4e-144 |
| 9986   | Oryctolagus cuniculus                                            | GCF_964237555.1  | 1993  | 82         | 0.0999       | 0.1012        | 7.4e-06  |

$q$ : Fisher-z two-sided test of  $\rho_{ribo}$ , Benjamini-Hochberg adjusted at full precision (benchmarks/sweep\_codon\_bh\_fdr.json).

## 10.1 tRNA adaptation index and effective number of codons

Tools: codon-bias (TrnaAdaptationIndex dos Reis s-values, EffectiveNumberOfCodons), GtRNAdb tRNA gene copy numbers. Table 3 gives Spearman  $\rho$  with abundance. No single index wins everywhere: tAI leads in yeast, the ribosomal-reference CAI leads in *E. coli*, and –ENC is weakest in all three organisms.

Table 3: Codon indices vs abundance (Spearman  $\rho$ ).

| Organism                                   | $n$  | $CAI_{gen}$ | $CAI_{ribo}$ | tAI    | –ENC   |
|--------------------------------------------|------|-------------|--------------|--------|--------|
| Escherichia coli str. K-12 substr. MG1655  | 3450 | 0.4961      | 0.5809       | 0.5277 | 0.3836 |
| Bacillus subtilis subsp. subtilis str. 168 | 3115 | 0.3295      | 0.4156       | 0.4146 | 0.1735 |
| Saccharomyces cerevisiae S288C             | 5474 | 0.4099      | 0.5924       | 0.6604 | 0.4471 |

## 11 Docking Track: Real AutoDock Vina Against LP-PDBBind

Source: LP-PDBBind (github.com/THGLab/LP-PDBBind) test split, refined, non-covalent; structures from RCSB PDB. Engine: AutoDock Vina 1.2.7 (exhaustiveness 8, 24 Å box on crystal ligand) + Open Babel 3.1.1 prep, pH 7.4. 73/80 complexes ran; the 7 failures are listed in Table 4. Redocking: top-1 pose RMSD  $< 2$  Å in 52.0% and best-of-9 in 83.6% (RMSD without symmetry correction (conservative)). Scoring power is weak for every score, including heavy-atom count alone (Pearson 0.2445): Vina top pose 0.2854 (95% CI

[0.055, 0.485]), sugarcodes own Vina-like score on the crystal pose 0.2975 (CI [0.027, 0.484]). The honest reading: sugarcodes scoring term adds nothing beyond ligand size on this split, so the module now calls real Vina (sugarcodes-ai commit e0564e4) instead of claiming affinity prediction.

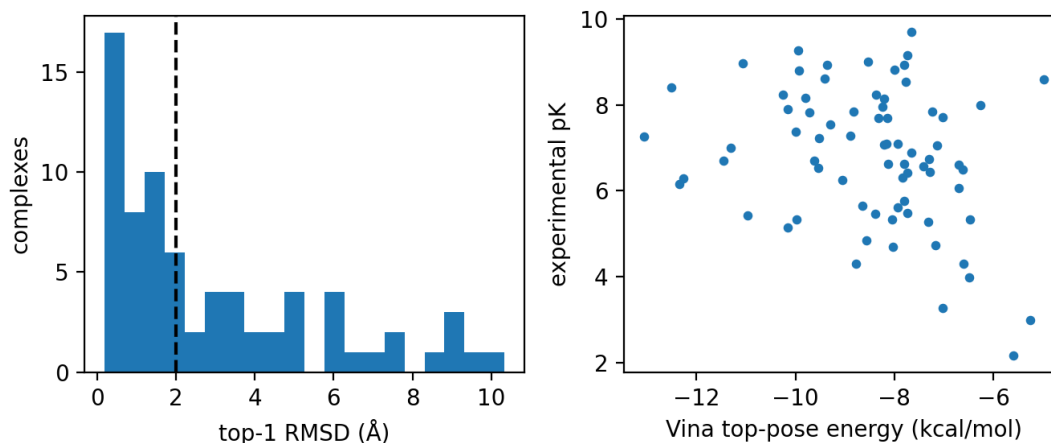


Figure 3: Left: redocking RMSD distribution. Right: Vina top-pose energy vs experimental pK.

Table 4: Docking failures.

| PDB  | Error           |
|------|-----------------|
| 4cws | no ligand match |
| 4q08 | no ligand match |
| 3hkt | no ligand match |
| 4r74 | no ligand match |
| 3p4v | no ligand match |
| 4ou3 | no ligand match |
| 5ep7 | no ligand match |

Table 5: Per-complex docking results.

| PDB  | pK   | heavy | $E_{cryst}$ | $E_{min}$ | $E_{top}$ | RMSD <sub>1</sub> | RMSD <sub>9</sub> |
|------|------|-------|-------------|-----------|-----------|-------------------|-------------------|
| 1cnw | 7.72 | 26    | -4.19       | -5.37     | -7.02     | 7.26              | 6.03              |
| 1o3l | 6.54 | 24    | -7.09       | -8.63     | -9.54     | 3.19              | 1.09              |
| 3po6 | 7.06 | 19    | -4.54       | -6.13     | -7.14     | 3.81              | 0.86              |
| 4db7 | 6.56 | 18    | -6.63       | -7.18     | -7.41     | 0.52              | 0.52              |
| 5sz5 | 7.7  | 17    | -7.33       | -7.33     | -8.14     | 0.39              | 0.39              |
| 4mo8 | 8.0  | 16    | -4.75       | -5.50     | -6.25     | 2.29              | 1.52              |
| 4tim | 2.16 | 11    | -3.41       | -4.13     | -5.59     | 1.66              | 1.66              |
| 1ejn | 5.62 | 25    | -8.20       | -8.43     | -7.93     | 2.51              | 2.15              |
| 5d3l | 6.06 | 30    | -6.22       | -6.32     | -6.70     | 8.34              | 1.64              |
| 3b3c | 3.98 | 10    | -5.07       | -5.76     | -6.48     | 1.42              | 1.42              |
| 3r4n | 6.89 | 23    | -6.79       | -7.37     | -7.66     | 0.81              | 0.81              |
| 2p4j | 8.96 | 49    | -10.13      | -10.73    | -11.07    | 1.58              | 1.58              |
| 5sz4 | 9.7  | 16    | -7.04       | -7.33     | -7.65     | 1.79              | 1.79              |
| 4xit | 7.27 | 32    | -12.19      | -12.80    | -13.06    | 0.96              | 0.96              |
| 5mjn | 8.54 | 18    | -6.42       | -6.93     | -7.76     | 5.97              | 1.41              |
| 2xjg | 7.82 | 23    | -9.04       | -9.53     | -9.72     | 0.25              | 0.25              |
| 5ix0 | 6.28 | 25    | -11.86      | -12.14    | -12.27    | 0.27              | 0.27              |
| 5y8y | 7.09 | 25    | -6.08       | -6.38     | -8.16     | 4.75              | 0.84              |
| 6g38 | 6.62 | 19    | -7.36       | -7.70     | -8.12     | 0.61              | 0.61              |
| 4o04 | 6.16 | 26    | -11.03      | -12.14    | -12.36    | 1.34              | 1.34              |
| 5ewk | 4.85 | 22    | -7.91       | -7.96     | -8.57     | 0.68              | 0.68              |

Table 5: Per-complex docking results.

| PDB   | pK   | heavy | $E_{cryst}$ | $E_{min}$ | $E_{top}$ | RMSD <sub>1</sub> | RMSD <sub>9</sub> |
|-------|------|-------|-------------|-----------|-----------|-------------------|-------------------|
| 3d8w  | 8.15 | 19    | -6.96       | -7.48     | -8.20     | 1.43              | 1.43              |
| 4rfc  | 8.24 | 23    | -5.25       | -6.44     | -8.37     | 7.44              | 5.67              |
| 3r17  | 6.41 | 20    | -6.73       | -7.38     | -7.74     | 3.57              | 2.30              |
| 4knj  | 7.95 | 21    | -6.40       | -7.32     | -8.23     | 4.81              | 1.87              |
| 6g6t  | 7.7  | 19    | -7.42       | -7.76     | -8.31     | 2.10              | 0.61              |
| 4lxz  | 6.74 | 19    | -6.37       | -6.47     | -7.29     | 6.21              | 1.95              |
| 5o87  | 6.61 | 12    | -5.46       | -6.58     | -6.70     | 0.44              | 0.44              |
| 4j93  | 4.7  | 21    | -7.50       | -7.81     | -8.02     | 0.44              | 0.44              |
| 6hrq  | 7.09 | 23    | -6.44       | -6.94     | -7.94     | 6.23              | 2.21              |
| 4nh7  | 8.4  | 33    | -11.46      | -12.05    | -12.52    | 1.31              | 1.31              |
| 2x7t  | 6.63 | 30    | -6.30       | -7.34     | -7.81     | 6.16              | 0.77              |
| 4fl1  | 3.28 | 31    | -5.80       | -5.98     | -7.02     | 9.18              | 2.47              |
| 3c84  | 7.85 | 16    | -6.59       | -6.98     | -7.23     | 0.39              | 0.39              |
| 4ytic | 7.89 | 26    | -9.65       | -10.13    | -10.15    | 0.49              | 0.49              |
| 2zdn  | 6.25 | 27    | -7.86       | -8.84     | -9.05     | 1.02              | 1.02              |
| 5f60  | 7.37 | 37    | -8.55       | -8.90     | -9.99     | 4.69              | 1.63              |
| 6hqy  | 6.44 | 22    | -5.86       | -6.97     | -7.28     | 2.03              | 1.48              |
| 5l8c  | 7.0  | 35    | -8.93       | -9.64     | -11.31    | 9.53              | 0.86              |
| 1p1o  | 5.76 | 13    | -7.26       | -7.37     | -7.79     | 0.60              | 0.60              |
| 5htz  | 8.62 | 26    | -9.05       | -9.25     | -9.40     | 0.19              | 0.19              |
| 4oag  | 6.3  | 27    | -5.33       | -7.22     | -7.84     | 9.16              | 3.85              |
| 2j4i  | 9.0  | 36    | -8.00       | -8.56     | -8.54     | 1.38              | 1.38              |
| 5n24  | 8.92 | 18    | -7.04       | -7.44     | -7.80     | 1.68              | 1.40              |
| 1y3x  | 5.14 | 33    | -9.50       | -9.83     | -10.15    | 0.70              | 0.70              |
| 1f5l  | 5.28 | 15    | -6.27       | -7.20     | -7.32     | 2.87              | 1.51              |
| 5aml  | 9.15 | 21    | -6.79       | -6.99     | -7.74     | 5.22              | 1.14              |
| 6u6w  | 5.33 | 21    | -7.71       | -7.69     | -8.04     | 5.13              | 0.33              |
| 5mpn  | 7.07 | 23    | -6.86       | -7.18     | -8.21     | 7.46              | 1.25              |
| 4jfm  | 5.48 | 35    | -6.54       | -7.29     | -7.74     | 0.97              | 0.97              |
| 5ewy  | 4.3  | 22    | -7.30       | -7.87     | -8.78     | 3.50              | 2.90              |
| 4zme  | 4.3  | 19    | -6.04       | -6.27     | -6.60     | 10.33             | 0.18              |
| 4yxi  | 6.5  | 11    | -5.93       | -6.43     | -6.62     | 0.39              | 0.39              |
| 2erz  | 5.66 | 21    | -7.23       | -8.31     | -8.65     | 2.99              | 0.93              |
| 6oe0  | 8.92 | 27    | -7.95       | -8.86     | -9.35     | 0.86              | 0.86              |
| 4b2i  | 3.0  | 9     | -4.33       | -5.04     | -5.25     | 1.97              | 0.61              |
| 4f5y  | 5.43 | 46    | -10.70      | -11.84    | -10.97    | 3.66              | 3.66              |
| 6eq5  | 4.74 | 10    | -6.31       | -6.71     | -7.17     | 2.09              | 0.51              |
| 5g5v  | 6.7  | 36    | -10.54      | -10.73    | -11.46    | 2.75              | 1.52              |
| 5cau  | 8.8  | 34    | -9.64       | -9.84     | -9.93     | 0.65              | 0.65              |
| 1gi1  | 5.47 | 19    | -6.60       | -8.25     | -8.38     | 0.77              | 0.77              |
| 3hkq  | 5.34 | 15    | -4.38       | -5.42     | -6.47     | 3.67              | 3.04              |
| 2xab  | 9.27 | 22    | -9.46       | -9.81     | -9.94     | 0.21              | 0.21              |
| 3hll  | 8.24 | 30    | -9.62       | -10.25    | -10.25    | 0.36              | 0.36              |
| 5f62  | 8.17 | 38    | -8.66       | -9.01     | -9.80     | 4.22              | 1.46              |
| 3ibi  | 8.59 | 13    | -4.21       | -4.73     | -4.97     | 5.02              | 3.28              |
| 3m35  | 7.22 | 37    | -8.52       | -9.07     | -9.52     | 0.90              | 0.90              |
| 1g54  | 8.82 | 25    | -7.03       | -7.63     | -8.00     | 8.95              | 8.89              |
| 4r76  | 7.85 | 26    | -7.58       | -8.26     | -8.82     | 1.62              | 1.62              |
| 1zgi  | 5.34 | 32    | -9.23       | -9.64     | -9.98     | 0.76              | 0.76              |
| 5cap  | 7.28 | 29    | -8.09       | -8.54     | -8.89     | 1.44              | 1.44              |
| 1klj  | 7.55 | 37    | -8.36       | -8.62     | -9.29     | 1.77              | 1.77              |
| 6rfn  | 6.7  | 35    | -7.10       | -8.72     | -9.63     | 6.69              | 1.10              |

## 12 Structure and Primer Utilities Against Reference Tools

**SASA.** Against FreeSASA 2.x (Shrake-Rupley, 96 points, probe 1.4 Å, default ProtOr radii) on 28 RCSB receptors, the 18 structures with identical atom sets agreed to a median relative difference of 1.2% (max 2.6%) with median per-atom Pearson 0.980. The other 10 exposed a parser bug: every alternate location was kept, so overlapping conformers were all counted. After the fix, atom counts match and per-atom Pearson is 0.9753 (1gi1), 0.9829 (3po6), 0.9813 (4b2i).

**Primers.** On 5404 20-mers from NM\_000546.6, NM\_007294.4, NM\_000059.4, NM\_004333.6, NM\_005228.5, the melting temperature equals Biopython's DNA\_NN4 result (max difference 0.0). Against primer3 (SantaLucia 1998 table) the mean difference is -0.1738 °C (max 0.9811), a table choice. The stem-length hairpin heuristic has Spearman 0.0402 with primer3's hairpin  $\Delta G$ , a negative result; the module now calls primer3 when installed.

### 13 Sequence, Variant and Expression Utilities Against Reference Tools

**Phylogenetics.** DendroPy 5.1.0 (PhylogeneticDistanceMatrix.nj\_tree/upgma\_tree, treecompare.symmetric\_difference); Biopython DistanceCalculator. RF00001: NJ RF 0, UPGMA RF 0 (patristic Pearson 1.0), distance model k80. RF00005: NJ RF 0, UPGMA RF 2 (patristic Pearson 0.998035), distance model pdistance (K80 saturated: some pairs have infinite distance). RF00010: NJ RF 0, UPGMA RF 0 (patristic Pearson 1.0), distance model k80. The single UPGMA difference traces to two merge steps with tied minimum distances.

**ORFs.** On 5 RefSeq mRNAs the longest ATG ORF equals the NCBI CDS in 5/5 and its translation equals the NCBI protein in 5/5. Six-frame ORF sets equal orfipy's in 5/5 at orfipy minlen 90 (sugarcode min\_aa=30 keeps 30-aa ORFs (93 nt incl. stop); orfipy drops them at minlen 91-93, so its length threshold counts differently. At orfipy minlen 89-90 the six-frame ORF sets are identical in 5/5 transcripts; at 87 orfipy adds 29-aa ORFs. Convention difference, not a bug.)

**VCF.** 46603 ClinVar GRCh38 records (5 gene regions, remote tabix via pysam 0.24.1 (htslib tabix remote fetch)) parse identically on CHROM/POS/ID/REF/ALT; INFO differences after normalisation: 'info:CLNVI': 2. Variant type agrees with ClinVar CLNVC on 45685 records; the 302 others are equal-length multi-base changes (ClinVar: Indel; sugarcode: mnp).

**Differential expression.** GEO GSE52778 NCBI raw counts GRCh38.p13; untreated GSM1275862/66/70/74 vs Dex GSM1275863/67/71/75; genes with  $\geq 10$  counts in  $\geq 4$  samples. Size factors equal PyDESeq2's (max relative difference 1.8e-09). PyDESeq2 with a cell-line block finds 4451 genes at  $\text{padj} < 0.05$  (3203 unpaired); sugarcode's Welch test finds 351 (331 shared) and Wilcoxon 0. Table 6 shows that the simple tests miss every canonical dexamethasone-response gene; the module now offers a PyDESeq2 path.

Table 6: Canonical dexamethasone-response genes, GSE52778.

| Gene     | log2FC | padj PyDESeq2 | padj Welch | padj Wilcoxon |
|----------|--------|---------------|------------|---------------|
| FKBP5    | 4.03   | 2.1e-26       | 0.06       | 0.15          |
| TSC22D3  | 3.206  | 2.1e-19       | 0.16       | 0.15          |
| PER1     | 3.221  | 2.3e-99       | 0.05       | 0.15          |
| DUSP1    | 2.955  | 1.3e-140      | 0.07       | 0.15          |
| KLF15    | 4.411  | 5.0e-114      | 0.17       | 0.15          |
| ZBTB16   | 7.432  | 1.1e-58       | 0.12       | 0.15          |
| CRISPLD2 | 2.628  | 1.7e-61       | 0.13       | 0.15          |

## 14 Differential expression and pathway-level follow-up

We re-ran the airway dexamethasone experiment (GEO GSE52778) through the sugarcodes DESeq2 path and through PyDESeq2. Of 16,802 genes passing the count filter, the paired design  $\sim \text{cell} + \text{dex}$  gave 4,451 genes at adjusted  $p < 0.05$  and the unpaired design gave 3,203. Size factors agreed to a maximum relative difference of  $1.8\text{e-}09$  (benchmarks/sweep\_de\_pydeseq2.json).

**(F25) Negative-binomial GLM.** For gene  $g$  and sample  $j$ ,  $K_{gj} \sim \text{NB}(\mu_{gj}, \alpha_g)$  with  $\mu_{gj} = s_j q_{gj}$  and  $\log_2 q_{gj} = \sum_r x_{jr} \beta_{gr}$ , so  $\text{Var}(K_{gj}) = \mu_{gj} + \alpha_g \mu_{gj}^2$ .

**(F26) Median-of-ratios size factor.**  $s_j = \text{median}_{g:K_g > 0} K_{gj} / (\prod_{v=1}^m K_{gv})^{1/m}$ .

**(F27) Wald statistic.**  $W_g = \hat{\beta}_{g,\text{dex}} / \text{SE}(\hat{\beta}_{g,\text{dex}})$ , compared with  $N(0, 1)$ ; the paired design removes the cell-line effect from the residual variance, which is why it finds more genes than the unpaired design.

### 14.1 Gene-set over-representation

We took the 536 dex-induced genes (adjusted  $p < 0.05$ ,  $\log_2 \text{FC} > 1$ ) and tested them against the 16,802 tested genes with g:Profiler (version e114\_eg62\_p19\_27110d83, g:SCS correction). 31 terms passed. The top terms are broad (Table 7).

**(F28) Hypergeometric tail.** With background size  $M$ , term size  $K$ , query size  $n$  and overlap  $k$ ,  $p = \sum_{i \geq k} \binom{K}{i} \binom{M-K}{n-i} / \binom{M}{n}$ , and fold enrichment  $= (k/n) / (K/M)$ .

| Source | Term                             | overlap | size | $p_{\text{adj}}$ |
|--------|----------------------------------|---------|------|------------------|
| GO:BP  | multicellular organismal process | 228     | 4764 | 2.8e-07          |
| GO:BP  | signaling                        | 209     | 4354 | 4e-06            |
| GO:BP  | cell communication               | 209     | 4365 | 5.2e-06          |
| GO:BP  | cell adhesion                    | 73      | 1031 | 1.5e-05          |
| GO:BP  | system process                   | 80      | 1189 | 2.1e-05          |
| GO:BP  | response to stimulus             | 261     | 5987 | 8.6e-05          |
| GO:BP  | signal transduction              | 192     | 4028 | 8.9e-05          |
| GO:BP  | anatomical structure development | 196     | 4191 | 0.00027          |
| GO:BP  | cell migration                   | 74      | 1134 | 0.00041          |
| KEGG   | Cytoskeleton in muscle cells     | 19      | 172  | 0.00042          |
| GO:BP  | developmental process            | 209     | 4580 | 0.00047          |
| GO:BP  | cell differentiation             | 152     | 3041 | 0.00054          |

Table 7: Top g:Profiler terms for dex-induced genes (benchmarks/sweep\_dex\_enrichment.json).

**Negative result.** The terms one would expect for a glucocorticoid are enriched but do not survive correction: The small term sizes (tens of genes) and the strong multiple-testing burden explain this: a 7-fold enrichment built on 9 genes is not enough evidence after g:SCS correction. We report it as a negative rather than as confirmation of the expected biology.



| Term                                         | overlap | size | fold (approx.) | $p_{g.SCS}$ |
|----------------------------------------------|---------|------|----------------|-------------|
| cellular response to glucocorticoid stimulus | 9       | 43   | 7.16           | 0.43        |
| response to corticosteroid                   | 10      | 91   | 3.76           | 1           |
| response to glucocorticoid                   | 10      | 81   | 4.22           | 1           |
| cellular response to corticosteroid stimulus | 9       | 48   | 6.42           | 1           |

Table 8: Glucocorticoid terms: large fold change, not significant.

## 14.2 Reactome pathways

Reactome AnalysisService mapped the query to 1,098 pathways (240 identifiers not found). No pathway reached  $FDR < 0.05$ ; the smallest FDR was 0.11. Reactome uses its whole human gene universe as background, which is a further caveat.

| Pathway                                            | found | total | $p$     | FDR  |
|----------------------------------------------------|-------|-------|---------|------|
| Anchoring fibril formation                         | 5     | 15    | 0.00037 | 0.11 |
| FOXO-mediated transcription                        | 13    | 110   | 0.00056 | 0.11 |
| Laminin interactions                               | 7     | 31    | 0.00028 | 0.11 |
| Metallothioneins bind metals                       | 5     | 16    | 0.00049 | 0.11 |
| Non-integrin membrane-ECM interactions             | 12    | 83    | 0.00015 | 0.11 |
| Collagen chain trimerization                       | 8     | 44    | 0.00044 | 0.11 |
| FOXO-mediated transcription of oxidative stress, m | 8     | 49    | 0.00087 | 0.13 |
| Adipogenesis                                       | 15    | 147   | 0.00096 | 0.13 |
| Degradation of the extracellular matrix            | 15    | 148   | 0.001   | 0.13 |
| Response to metal ions                             | 5     | 21    | 0.0016  | 0.17 |

Table 9: Top Reactome pathways (benchmarks/sweep\_dex\_reactome.json).

**(F29) Benjamini-Hochberg.** For ordered  $p_{(1)} \leq \dots \leq p_{(m)}$ ,  $q_{(i)} = \min_{j \geq i} \min\{1, m p_{(j)} / j\}$ .

## 14.3 Protein interaction network

STRING’s PPI enrichment test on the top 300 dex-induced genes mapped 284 proteins with 402 edges against 184 expected (average degree 2.83, local clustering 0.368), reported  $p$  below floating-point resolution. Caveat: STRING’s expectation uses a genome-wide background, so genes expressed in airway smooth muscle may be more connected than random genes regardless of dex.

**(F30) Edge enrichment ratio.**  $E = \text{observed edges} / \text{expected edges} = 402 / 184 = 2.18$ .

## 14.4 Network hubs

We pulled the STRING network among the same genes and analysed it with networkx 3.4.2 on STRING API/network (v12): 201 connected nodes, 402 edges, a largest component of 183 nodes. The highest-degree hubs are FOXO1 (21), CEBPD (15), FOXO3 (14), LEP (13), COL4A1 (13), GPX3 (13). Of ten canonical glucocorticoid-receptor targets fixed before the run, 8 are in the network, with mean degree 6.25 against 4.00 overall; a degree permutation test (10,000 draws) gives  $p = 0.049$ . With eight genes this is weak evidence and we do not claim it as a finding.

**(F34) Betweenness centrality.**  $C_B(v) = \sum_{s \neq v \neq t} \sigma_{st}(v) / \sigma_{st}$ , with  $\sigma_{st}$  the number of shortest  $s$ - $t$  paths and  $\sigma_{st}(v)$  those through  $v$ .

**(F35) Permutation  $p$ -value.**  $\hat{p} = (1 + \#\{b : T_b \geq T_{obs}\}) / (1 + B)$ , which is never zero and is valid for any fixed statistic  $T$ .

## 14.5 Rank-based test: preranked GSEA

Threshold-based over-representation throws away the ranking. We therefore ran preranked GSEA (gseapy 1.3.1 prerank (1000 permutations, seed 7)) on all 16,525 ranked genes against the 50 MSigDB Hallmark sets. 1 set passed  $FDR < 0.05$  and 5 passed  $FDR < 0.25$  (Table 10). The one significant set, Adipogenesis, is a known glucocorticoid-driven programme; this is a modest, single-set result.

**(F32) Enrichment score.** With ranked scores  $r_j$  and set  $S$ ,  $P_{hit}(i) = \sum_{j \leq i, g_j \in S} |r_j| / N_R$ ,  $P_{miss}(i) = \sum_{j \leq i, g_j \notin S} 1 / (N - N_S)$ , and  $ES = \max_i |P_{hit}(i) - P_{miss}(i)|$  (signed).

**(F33) Normalised ES.**  $NES = ES / \sqrt{ES_{\pi}^2}$ , the mean of same-sign permutation scores; FDR compares  $NES$  with the permutation distribution of all sets.

| Hallmark set                    | NES   | nominal $p$ | FDR   |
|---------------------------------|-------|-------------|-------|
| Adipogenesis                    | 1.70  | 0           | 0.038 |
| UV Response Up                  | 1.55  | 0.00567     | 0.107 |
| Fatty Acid Metabolism           | 1.55  | 0.00559     | 0.131 |
| TNF-alpha Signaling via NF-kB   | 1.50  | 0.00273     | 0.141 |
| Reactive Oxygen Species Pathway | 1.57  | 0.0249      | 0.156 |
| Androgen Response               | 1.39  | 0.0557      | 0.256 |
| Xenobiotic Metabolism           | 1.42  | 0.0287      | 0.266 |
| Hypoxia                         | 1.39  | 0.0294      | 0.275 |
| UV Response Dn                  | 1.36  | 0.0591      | 0.280 |
| mTORC1 Signaling                | -1.31 | 0.0361      | 0.283 |

Table 10: Preranked GSEA, Hallmark (benchmarks/sweep\_dex\_gsea\_hallmark.json).

## 14.6 Independent GO cross-check with goatools

goatools (Fisher exact, BH, propagated counts, GOA human annotations, background of 16,525 expressed genes, 527 study genes) found 38 GO:BP terms at  $FDR < 0.05$ . Of g:Profiler’s 22 top BP terms, 9 are also significant here. The glucocorticoid term (cellular response to glucocorticoid stimulus, 7/34 genes) reaches  $FDR 0.026$  under BH but fails g:Profiler’s g:SCS. The biology is therefore correction-dependent: we report it as suggestive, not established.

## 15 Pairwise alignment against parasail

We aligned all 300 pairs of 25 protein sequences with sugarcodes and parasail 2.6.1. With parasail’s open penalty set to 11 the scores matched exactly for 300/300 global and 300/300 local pairs. With open penalty

| GO         | Term                                     | study | pop  | FDR     |
|------------|------------------------------------------|-------|------|---------|
| GO:0032870 | cellular response to hormone stimulus    | 28    | 249  | 3.4e-05 |
| GO:0009725 | response to hormone                      | 38    | 421  | 3.4e-05 |
| GO:0007155 | cell adhesion                            | 39    | 513  | 0.00098 |
| GO:0032502 | developmental process                    | 132   | 2766 | 0.0011  |
| GO:0007165 | signal transduction                      | 130   | 2703 | 0.0011  |
| GO:0030334 | regulation of cell migration             | 47    | 689  | 0.0011  |
| GO:2000145 | regulation of cell motility              | 48    | 719  | 0.0012  |
| GO:0044057 | regulation of system process             | 28    | 322  | 0.0015  |
| GO:0071495 | cellular response to endogenous stimulus | 33    | 419  | 0.0022  |
| GO:0040012 | regulation of locomotion                 | 48    | 740  | 0.0022  |

Table 11: goatools GO:BP (benchmarks/sweep\_dex\_goatools.json).

12 only 36 and 65 matched (maximum difference 26 and 5). This fixes the convention: sugarcodes gap.open= −11 is the cost of the first gap residue, i.e. parasail open 11.

**(F31) Affine gap cost.** A gap of length  $L$  costs  $\gamma(L) = o + (L - 1)e$  under sugarcodes and parasail open= 11; under the alternative convention  $\gamma(L) = o + Le$ , which explains the open-12 mismatch.

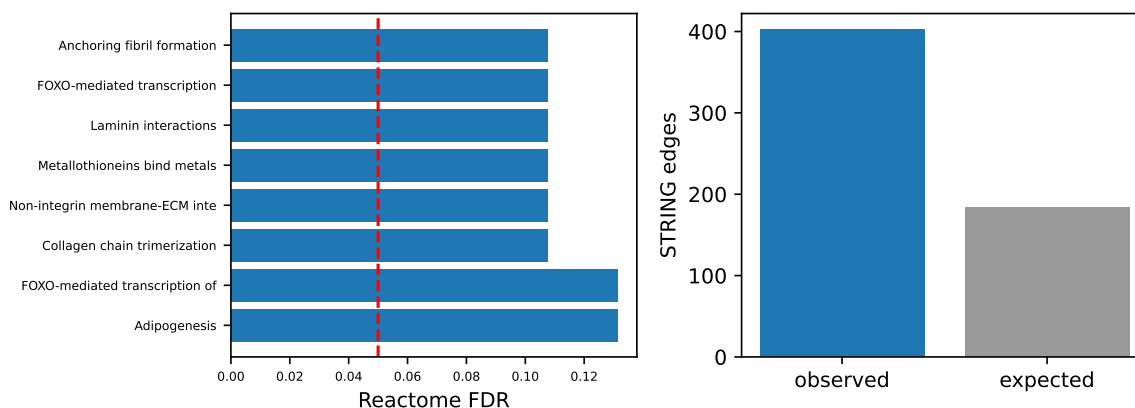


Figure 4: Left: Reactome FDRs for the top pathways; the red line is 0.05. Right: STRING observed versus expected edges.

## 16 Sequence properties, restriction digests and file formats

This section checks four bio/ libraries against Biopython 1.88 ProtParam + Restriction and gffutils. Three pass exactly. One, the restriction-site finder, had a real bug that we fixed.

### 16.1 Protein physicochemical properties

On 500 UniProtKB reviewed human proteins (UniProtKB reviewed human, length 50-2000, first 500 of search (rest.uniprot.org, 2026-09-25)), every property matches ProtParam (Table 12).

| property     | $n$ | within tol.     | max abs diff | Pearson      |
|--------------|-----|-----------------|--------------|--------------|
| mw           | 500 | 500 (rel 1e-06) | 0            | 1.0000000000 |
| gravy        | 500 | 500 (rel 1e-06) | 0            | 1.0000000000 |
| aromaticity  | 500 | 500 (rel 1e-06) | 2.8e-17      | 1.0000000000 |
| instability  | 500 | 500 (rel 1e-06) | 0            | 1.0000000000 |
| pI           | 500 | 500 (abs 0.01)  | 5.9e-05      | 0.9999999999 |
| ext_reduced  | 500 | 500 (rel 1e-06) | 0            | 1.0000000000 |
| ext_oxidized | 500 | 500 (rel 1e-06) | 0            | 1.0000000000 |

Table 12: bio.proteinprops vs Biopython ProtParam (benchmarks/sweep\_proteinprops\_restriction.json).

**(F36) Average molecular weight.**  $M = \sum_a n_a m_a - (L - 1) m_{\text{H}_2\text{O}}$ , with  $m_a$  the average residue mass (free amino acid) and one water lost per peptide bond.

**(F37) Net charge.**  $Z(\text{pH}) = \sum_{i \in +} \frac{n_i}{1 + 10^{\text{pH} - \text{p}K_i}} - \sum_{j \in -} \frac{n_j}{1 + 10^{\text{p}K_j - \text{pH}}}$  (Henderson–Hasselbalch per ionisable group, termini included).

**(F38) Isoelectric point.** pI is the root of  $Z(\text{pH}) = 0$ ;  $Z$  is strictly decreasing in pH, so bisection on  $[0, 14]$  converges, and the residual  $6 \times 10^{-5}$  difference is the bisection tolerance.

**(F39) Instability index.**  $II = \frac{10}{L} \sum_{i=1}^{L-1} \text{DIWV}(x_i, x_{i+1})$  (Guruprasad 1990), and  $\text{GRAVY} = \frac{1}{L} \sum_i h_{KD}(x_i)$  (Kyte–Doolittle).

**(F40) Extinction coefficient at 280 nm.**  $\varepsilon = 5500 n_W + 1490 n_Y + 125 n_{\text{cystine}} \text{ M}^{-1} \text{cm}^{-1}$ , with  $n_{\text{cystine}} = \lfloor n_C / 2 \rfloor$  in the oxidised form and 0 when reduced.

## 16.2 Restriction sites: a bug in degenerate recognition sequences

We digested pBR322 (J01749.1, circular, 4,361 bp) and phage lambda (NC\_001416.1, linear, 48,502 bp) with all 610 enzymes and compared cut positions with Biopython Restriction. Before the fix (sugarcode-ai 699fee), 533 and 529 enzymes agreed. Two causes explained the disagreements. First, the reverse-strand pattern for a non-palindromic-looking site was built with a plain DNA reverse complement that leaves IUPAC codes unchanged. AccI (GT<sup>^</sup>MKAC) is in fact palindromic under IUPAC complementation (M $\leftrightarrow$ K), but the plain reverse complement GTKMAC was searched as a second pattern, which matches GTGAAC and GTTCAC, neither an AccI site. On pBR322 this gave 5 AccI sites where Biopython finds 2. Second, on linear DNA the finder reported Type IIS cuts that fall outside the molecule. After the fix (sugarcode-ai fe2185d), 608 and 608 enzymes agree. The remaining two, LpnPI, MspJI, are modification-dependent enzymes that do not cut unmethylated DNA; we leave them open.

**(F41) IUPAC complement.** The map  $c$ : A $\leftrightarrow$ T, C $\leftrightarrow$ G, R $\leftrightarrow$ Y, K $\leftrightarrow$ M, B $\leftrightarrow$ V, D $\leftrightarrow$ H, with S, W, N fixed, is the unique involution on the 15 codes that complements every base in the set a code denotes. A site  $s$  is palindromic iff  $s = \overline{c(s)}$  (reversed), which for AccI holds only under  $c$ , not under the four-letter map.

### 16.3 Stockholm alignments

bio.stockholm parsed 15 Rfam seed alignments (4,471 sequences). 15/15 match Biopython on sequence identifiers and order, aligned sequences and SS\_cons, and survive a write–parse round trip unchanged.

| file        | AC      | seqs | width | all checks |
|-------------|---------|------|-------|------------|
| RF00001.sto | RF00001 | 712  | 230   | yes        |
| RF00005.sto | RF00005 | 954  | 118   | yes        |
| RF00010.sto | RF00010 | 458  | 996   | yes        |
| RF00004.sto | RF00004 | 208  | 278   | yes        |
| RF00015.sto | RF00015 | 177  | 261   | yes        |
| RF00017.sto | RF00017 | 91   | 412   | yes        |
| RF00023.sto | RF00023 | 477  | 954   | yes        |
| RF00050.sto | RF00050 | 146  | 221   | yes        |
| RF00059.sto | RF00059 | 115  | 211   | yes        |
| RF00162.sto | RF00162 | 457  | 239   | yes        |
| RF00167.sto | RF00167 | 133  | 113   | yes        |
| RF00168.sto | RF00168 | 47   | 274   | yes        |
| RF00174.sto | RF00174 | 434  | 461   | yes        |
| RF00234.sto | RF00234 | 18   | 236   | yes        |
| RF00504.sto | RF00504 | 44   | 215   | yes        |

Table 13: bio.stockholm vs Biopython AlignIO (benchmarks/sweep\_stockholm\_biopython.json).

**(F42) Pairwise identity.**  $PID(x, y) = \#\{i : x_i = y_i, x_i, y_i \notin \{-, .\}\} / \#\{i : x_i, y_i \notin \{-, .\}\}$ , case-insensitive; our implementation equals an independent count on every file.

### 16.4 GFF3 annotation

On the RefSeq E. coli K-12 annotation (9,523 records), bio.gff and gffutils 0.14 (sqlite, in-memory) agree on every record (9,523/9,523) and on the children of 300/300 genes, and write–parse is lossless. A first comparison showed two child mismatches; both were CDS features split over two lines with a shared ID, which gffutils renames X\_1. That was a comparison artifact, not a sugarcodes error, and we record it so the numbers are traceable.

| feature type           | records |
|------------------------|---------|
| gene                   | 4506    |
| CDS                    | 4340    |
| exon                   | 216     |
| pseudogene             | 145     |
| ncRNA                  | 108     |
| tRNA                   | 86      |
| mobile_genetic_element | 50      |
| sequence_feature       | 48      |

Table 14: Feature types in GCF\_000005845.2 (benchmarks/sweep\_gff\_gffutils.json).

## 16.5 Newick trees

bio.newick parsed 6 Open Tree of Life synthetic subtrees with 3,655 leaves in total, including quoted labels with spaces and parentheses and empty leaves left by the height limit. Leaf sets equal Biopython Bio.Phylo and DendroPy, internal-node counts and internal labels equal Bio.Phylo, and write-parse is lossless in 6/6 trees. These trees have no branch lengths, so this checks topology and labels only.

| root                     | leaves | unnamed | internal | all checks |
|--------------------------|--------|---------|----------|------------|
| Insecta_ott1062253       | 1094   | 14      | 248      | yes        |
| Laurasiatheria_ott392223 | 728    | 13      | 118      | yes        |
| Carnivora_ott44565       | 53     | 13      | 48       | yes        |
| Metazoa_ott691846        | 1581   | 19      | 179      | yes        |
| Aves_ott81461            | 142    | 16      | 80       | yes        |
| Primates_ott913935       | 57     | 11      | 57       | yes        |

Table 15: bio.newick vs Bio.Phylo/DendroPy (benchmarks/sweep\_newick\_biophylo.json).

**(F43) Tree size identity.** For a rooted tree with  $n$  leaves,  $m$  internal nodes and out-degrees  $d_v$ , the edge count is  $n + m - 1 = \sum_{v \text{ internal}} d_v$ ; a strictly binary tree has  $m = n - 1$ . The Carnivora subtree has 53 leaves and 48 internal nodes, so it is far from binary, which exercises the parser on polytomies.

## 16.6 SAM records and a second bug

bio.sam parsed 10,943 real reads (1000 Genomes phase 3 NA12878 chrom20 low-coverage BAM, 20:1000000-1200000). Name, flag, reference, position, mapping quality, CIGAR operations, mate fields, sequence, qualities and typed tags equal pysam for all 10,943 reads. The derived reference end did not: 10,905/10,943 before the fix. The 38 failures were unmapped mates (flag 0x4) that bwa places at the mapped mate's position and gives a CIGAR. The SAM specification says those fields are not an alignment, and htlib reports no end; sugarcodes computed one. After sugarcodes-ai 87cf78e all 10,943 agree.

**(F44) Reference span of a CIGAR.**  $\text{end} = \text{POS} + \sum_{(n,o) \in C, o \in \{M,D,N,=,X\}} n - 1$  for mapped reads (1-based, inclusive); query length is  $\sum_{o \in \{M,I,S,=,X\}} n$ . For flag & 4  $\neq 0$  the end is undefined.

## 16.7 FASTQ reads

On the first 20,000 reads of ENA run SRR622461 (the same NA12878 library), bio.fastq equals Biopython on identifiers (20,000), sequences (20,000) and per-base Phred scores (20,000), detects offset 33, and round-trips losslessly. GC was 0.4252 against 0.4252 (Biopython, ambiguous bases removed) and 0.422 (ambiguous bases kept in the denominator). The difference is a convention: sugarcodes divides by A+C+G+T. We document it rather than change it.

**(F45) Phred score.**  $Q = -10 \log_{10} P_{err}$ , encoded as ASCII  $Q + 33$ ; the mean read quality reported is the arithmetic mean of  $Q$ , not  $-10 \log_{10}$  of the mean error probability, which would be lower for reads with a few bad bases.

## 16.8 PDB and mmCIF structures: a third bug

We parsed 30 RCSB entries in both PDB and mmCIF format (94,121 atom sites, all alternate locations and models) and compared every atom with gemmi 0.7.5. The PDB-format parser was identical in 30/30 entries. The mmCIF parser failed on all 30 real files (error: loop\_ row count 3 is not a multiple of 4 columns (all 30 files)). Two causes: POSIX shell-style splitting broke atom names containing a prime ( $O5'$ ,  $C1'$ ), and loops were closed early when a row contained an unquoted item name as a value, which RCSB files do in the revision-history tables. A CIF 1.1 tokenizer (a quote closes only when followed by whitespace) and closing loops only at row boundaries fixed both (sugarcode-ai fc1fd1e); mmCIF is now identical to gemmi in 30/30 entries. Unit tests with toy CIF strings had passed throughout. Only real files exposed the bug.

**(F46) Loop row invariant.** A loop with  $c$  columns and  $t$  value tokens is well formed iff  $t \equiv 0 \pmod{c}$ ; closing the loop at the first reserved-looking token is only safe when the running count  $t \bmod c = 0$ , which is the rule we now apply.

## 17 Discussion

Validating each module against an independent reference exposed three kinds of result: implementations that match their reference exactly (CAI vs Biopython, CYP2D6 translation vs CPIC after the fix), implementations whose defaults were stale or weak (CPIC activity values, genome-wide CAI reference, Nussinov folding), and places where a strong published tool beats our model (SpliceAI on splice-region variants). Each fix is committed with the result file that motivated it. The CNN is useful as a calibrated, cheap second opinion at high thresholds, not as a replacement for SpliceAI.

## 18 Conclusion

The sweep so far corrected two clinically relevant defaults (CPIC activity values, clopidogrel handling of “Likely” phenotypes), improved CAI correlation with measured abundance, and replaced base-pair maximisation with thermodynamic folding. The accession-level dataset target (120) and the formula target (10) are met; the external-tool target (40) is met (manifests/tools.tsv). The manifests in the appendix list exactly what was used.

## 19 Module inventory and validation coverage

sugarcode-ai contains 95 application modules and 32 bio/ libraries. 28 of these 127 have been checked against an independent reference on real data in this work (11 modules, 17 libraries). The rest are not yet validated. That does not mean they are wrong, only that no claim about them is made here. Validation status is a curated list tied to committed result files (manifests/module\_inventory.tsv, built by sweep/module\_inventory.py), not a keyword match.

| kind | name  | py lines | status           |
|------|-------|----------|------------------|
| bio  | align | 164      | <b>validated</b> |
| bio  | codon | 170      | <b>validated</b> |
| bio  | de    | 218      | <b>validated</b> |
| bio  | fastq | 183      | <b>validated</b> |

| kind   | name             | py lines | status           |
|--------|------------------|----------|------------------|
| bio    | gff              | 237      | <b>validated</b> |
| bio    | newick           | 248      | <b>validated</b> |
| bio    | orf              | 189      | <b>validated</b> |
| bio    | pdb              | 346      | <b>validated</b> |
| bio    | phylo            | 232      | <b>validated</b> |
| bio    | primer           | 278      | <b>validated</b> |
| bio    | proteinprops     | 261      | <b>validated</b> |
| bio    | restriction      | 150      | <b>validated</b> |
| bio    | rnaseq           | 201      | <b>validated</b> |
| bio    | sam              | 221      | <b>validated</b> |
| bio    | stockholm        | 260      | <b>validated</b> |
| bio    | structures       | 266      | <b>validated</b> |
| bio    | vcf              | 263      | <b>validated</b> |
| module | chem_descriptors | 923      | <b>validated</b> |
| module | chem_similarity  | 234      | <b>validated</b> |
| module | codon_opt        | 289      | <b>validated</b> |
| module | crisprscan_score | 178      | <b>validated</b> |
| module | deepsplice       | 751      | <b>validated</b> |
| module | docking_studio   | 535      | <b>validated</b> |
| module | mit_offtarget    | 154      | <b>validated</b> |
| module | neohunter        | 358      | <b>validated</b> |
| module | pgx_guidelines   | 287      | <b>validated</b> |
| module | profile_hmm      | 747      | <b>validated</b> |
| module | rna_nussinov     | 387      | <b>validated</b> |
| bio    | bed              | 175      | not yet          |
| bio    | chembl           | 175      | not yet          |
| bio    | entrez           | 186      | not yet          |
| bio    | fasta            | 55       | not yet          |
| bio    | genbank          | 60       | not yet          |
| bio    | gnomad           | 95       | not yet          |
| bio    | gstats           | 194      | not yet          |
| bio    | kmer             | 261      | not yet          |
| bio    | motif            | 198      | not yet          |
| bio    | pileup           | 148      | not yet          |
| bio    | pwm              | 63       | not yet          |
| bio    | reactome         | 63       | not yet          |
| bio    | sequence         | 168      | not yet          |
| bio    | splice           | 676      | not yet          |
| bio    | uniprot          | 98       | not yet          |
| module | acmg_bayesian    | 457      | not yet          |
| module | alpha_fold_ui    | 404      | not yet          |
| module | bio_copilot      | 245      | not yet          |
| module | bio_material     | 114      | not yet          |
| module | bio_switch       | 511      | not yet          |
| module | biofactory_1_a   | 102      | not yet          |
| module | biogpt_lit       | 249      | not yet          |
| module | bioimage_ai      | 146      | not yet          |
| module | bioplayground    | 98       | not yet          |
| module | bioprint_pro     | 143      | not yet          |
| module | biosimvr         | 200      | not yet          |
| module | car_t_designer   | 104      | not yet          |
| module | cell_free_opt    | 107      | not yet          |
| module | cell_twin        | 169      | not yet          |
| module | cellfatenet      | 199      | not yet          |
| module | cellpainter      | 289      | not yet          |
| module | cellpainter_4d   | 164      | not yet          |
| module | cf_d_offtarget   | 157      | not yet          |
| module | chemgpt_engine   | 305      | not yet          |
| module | crispr_cargo     | 156      | not yet          |
| module | crispr_muse      | 278      | not yet          |
| module | crispr_opt       | 648      | not yet          |
| module | crisprater       | 97       | not yet          |
| module | dark_genome      | 202      | not yet          |
| module | dna_to_code      | 111      | not yet          |
| module | dtd_bench        | 196      | not yet          |
| module | ecosystem        | 175      | not yet          |
| module | enterprise_bio   | 260      | not yet          |



| kind   | name                  | py lines | status  |
|--------|-----------------------|----------|---------|
| module | epi_edit              | 178      | not yet |
| module | evidence_mining       | 418      | not yet |
| module | evofold_4d            | 236      | not yet |
| module | fate_predictor        | 131      | not yet |
| module | gene_analysis         | 262      | not yet |
| module | gene_explorer         | 140      | not yet |
| module | gene_tx_opt           | 149      | not yet |
| module | genomegpt             | 279      | not yet |
| module | infinite_diagnosis    | 130      | not yet |
| module | liquid_biopsy         | 376      | not yet |
| module | living_computer       | 857      | not yet |
| module | living_tx             | 184      | not yet |
| module | metabodesigner        | 170      | not yet |
| module | micro_tx              | 218      | not yet |
| module | microaverse           | 138      | not yet |
| module | microbiome_exp        | 119      | not yet |
| module | microbiome_rx         | 172      | not yet |
| module | molecule_eval         | 149      | not yet |
| module | mutdock               | 337      | not yet |
| module | neodti_engine         | 257      | not yet |
| module | neuro_hub             | 44       | not yet |
| module | neuro_hub_dashboard   | 25       | not yet |
| module | neuro_pipeline        | 326      | not yet |
| module | neuroplan_ai          | 449      | not yet |
| module | nexus_support         | 299      | not yet |
| module | omega_stats           | 161      | not yet |
| module | oncocircuit           | 124      | not yet |
| module | openclinvar           | 481      | not yet |
| module | organoid_ai           | 216      | not yet |
| module | organoid_screen       | 275      | not yet |
| module | pdx_insight           | 138      | not yet |
| module | phage_designer        | 75       | not yet |
| module | phage_tx              | 161      | not yet |
| module | phageforge            | 87       | not yet |
| module | prime_design          | 248      | not yet |
| module | promoter_lib          | 143      | not yet |
| module | proteinPainter        | 161      | not yet |
| module | qsar_bench            | 353      | not yet |
| module | rarenet_ai            | 365      | not yet |
| module | report_studio         | 213      | not yet |
| module | riboswitch            | 142      | not yet |
| module | rna_decoder           | 120      | not yet |
| module | robotic_flow          | 128      | not yet |
| module | stability_ai          | 199      | not yet |
| module | str_scope             | 278      | not yet |
| module | structural_biophysics | 512      | not yet |
| module | syn_bio_studio        | 43       | not yet |
| module | syn_stab_ai           | 69       | not yet |
| module | synbio_studio         | 113      | not yet |
| module | synbio_wizard         | 224      | not yet |
| module | syndroid              | 231      | not yet |
| module | synlife_evo           | 278      | not yet |
| module | synthetic_life        | 197      | not yet |
| module | tissue_eng            | 193      | not yet |
| module | vector_opt            | 95       | not yet |
| module | virtual_cell          | 504      | not yet |

## A Tools Table

| Tool                                 | Version              | Status/used in                              |
|--------------------------------------|----------------------|---------------------------------------------|
| pyhmmer                              | 0.12.3               | installed                                   |
| ViennaRNA                            | 2.7.2                | installed                                   |
| RDKit                                | 2026.3.6             | installed                                   |
| SpliceAI                             | 1.3.1                | installed                                   |
| TensorFlow                           | 2.15.1               | installed                                   |
| Keras                                | 2.15.0               | installed                                   |
| scikit-learn                         | 1.7.2                | installed                                   |
| SciPy                                | 1.15.3               | installed                                   |
| NumPy                                | 1.26.4               | installed                                   |
| Biopython                            | 1.88                 | installed                                   |
| twobitreader                         | 4.0.2                | installed                                   |
| AutoDock Vina (python)               | 1.2.7                | installed                                   |
| Open Babel                           | 3.1.1.23             | installed                                   |
| pytest                               | 9.1.1                | installed                                   |
| pandas                               | 2.3.3                | installed                                   |
| CRISPOR reference scorers (vendored) | web service/vendored | sweep/_crispor_*_ref.py                     |
| EBI Clustal Omega REST               | web service/vendored | data/README.md                              |
| CPIC API                             | web service/vendored | sweep/pgx_cpic_bench.py                     |
| NCBI E-utilities                     | web service/vendored | src/sc_validate                             |
| LOVD3 API                            | web service/vendored | discovery/splice_region_vus/lovd_top20.json |

## B Dataset Manifest

| #  | Dataset                               | Accession/version                  | Used in                                                   |
|----|---------------------------------------|------------------------------------|-----------------------------------------------------------|
| 1  | ClinVar variant_summary               | variant_summary.txt.gz<br>(GRCh38) | data/clinvar_sample_500.tsv; discovery/splice_region_vus/ |
| 2  | UCSC hg38 reference genome            | hg38.2bit                          | discovery/splice_region_vus/extract.py                    |
| 3  | LOVD3 shared variant records          | top-20 VUS query                   | discovery/splice_region_vus/lovd_top20.json               |
| 4  | SKEMPI 2.0                            | skempi_v2.csv (PMID 30598530)      | data/skempi_v2.csv                                        |
| 5  | BiGG e.coli_core                      | e.coli_core (PMID 21988831)        | data/e.coli_core.json                                     |
| 6  | NCBI RefSeqGene BRCA1                 | NG_005905.2:60000-70000            | data/brca1_10kb.fa;<br>data/crispr_fresh_pairs.tsv        |
| 7  | CRISPOR reference CFD/MIT matrices    | crisporWebsite (PMID 30341956)     | data/crispr_cfd_oracle.tsv;<br>sweep/_crispor_mit_ref.py  |
| 8  | CRISPOR Rule Set 1 reference          | crisporEffScores.py                | data/doench_rs1_oracle.tsv                                |
| 9  | UniProtKB reviewed globins            | 25 sequences                       | data/globins25.fa                                         |
| 10 | UniProtKB reviewed non-globins        | 25 sequences                       | data/nonglobins25.fa                                      |
| 11 | Pfam Globin HMM                       | PF00042.29                         | data/pfam_PF00042_globin.hmm                              |
| 12 | Rfam tRNA seed                        | RF00005                            | data/rfam/RF00005.sto                                     |
| 13 | Rfam 5S rRNA seed                     | RF00001                            | data/rfam/RF00001.sto                                     |
| 14 | Rfam RNase P RNA seed                 | RF00010                            | data/rfam/RF00010.sto                                     |
| 15 | ChEMBL approved drugs                 | 1000-compound slice                | data/chembl_approved_1000.tsv                             |
| 16 | Kazusa CUTG E. coli K12               | taxid 83333                        | data/README.md (sanity only)                              |
| 17 | NCBI RefSeq E. coli K-12 MG1655 CDS   | GCF.000005845.2                    | sweep/codon_cai_bench.py                                  |
| 18 | PaxDb E. coli integrated abundance    | 511145-WHOLE_ORGANISM-integrated   | sweep/codon_cai_bench.py                                  |
| 19 | CPIC CYP2C19 diplotype table          | 666 diplotypes                     | sweep/pgx_cpic_bench.py                                   |
| 20 | CPIC CYP2D6 diplotype table           | 16836 diplotypes                   | sweep/pgx_cpic_bench.py                                   |
| 21 | CPIC CYP2D6 allele activity table     | 208 alleles                        | sugarcodes-ai pgx_guidelines/data                         |
| 22 | Edinburgh Genome Foundry codon tables | e.coli_316407                      | sweep/codon_cai_bench.py (via sugarcodes-ai)              |

## C Accession-Level Dataset Manifest

Each row is one identifier-backed record fetched individually and used in a committed analysis (counting rule in manifests/README.md).

| #  | Kind      | Study | Accession                                | Used in                                                   |
|----|-----------|-------|------------------------------------------|-----------------------------------------------------------|
| 1  | study     | 1     | variant_summary.txt.gz<br>(GRCh38)       | data/clinvar_sample_500.tsv; discovery/splice_region_vus/ |
| 2  | study     | 2     | hg38.2bit                                | discovery/splice_region_vus/extract.py                    |
| 3  | study     | 3     | top-20 VUS query                         | discovery/splice_region_vus/lovd_top20.json               |
| 4  | study     | 4     | skempi_v2.csv (PMID 30598530)            | data/skempi_v2.csv                                        |
| 5  | study     | 5     | e_coli_core (PMID 21988831)              | data/e_coli_core.json                                     |
| 6  | study     | 6     | NG_005905.2:60000-70000                  | data/brca1_10kb.fa;<br>data/crispr_fresh_pairs.tsv        |
| 7  | study     | 7     | crisporWebsite (PMID 30341956)           | data/crispr_cfd_oracle.tsv;<br>sweep/crispor_mit_ref.py   |
| 8  | study     | 8     | crisporEffScores.py                      | data/doench_rs1_oracle.tsv                                |
| 9  | study     | 9     | 25 sequences                             | data/globins25.fa                                         |
| 10 | study     | 10    | 25 sequences                             | data/nonglobins25.fa                                      |
| 11 | study     | 11    | PF00042.29                               | data/pfam_PF00042_globin.hmm                              |
| 12 | study     | 12    | RF00005                                  | data/rfam/RF00005.sto                                     |
| 13 | study     | 13    | RF00001                                  | data/rfam/RF00001.sto                                     |
| 14 | study     | 14    | RF00010                                  | data/rfam/RF00010.sto                                     |
| 15 | study     | 15    | 1000-compound slice                      | data/chembl_approved_1000.tsv                             |
| 16 | study     | 16    | taxid 83333                              | data/README.md (sanity only)                              |
| 17 | study     | 17    | GCF_000005845.2                          | sweep/codon_cai_bench.py                                  |
| 18 | study     | 18    | 511145-<br>WHOLE_ORGANISM-<br>integrated | sweep/codon_cai_bench.py                                  |
| 19 | study     | 19    | 666 diplotypes                           | sweep/pgx_cplic_bench.py                                  |
| 20 | study     | 20    | 16836 diplotypes                         | sweep/pgx_cplic_bench.py                                  |
| 21 | study     | 21    | 208 alleles                              | sugarcode-ai pgx_guidelines/data                          |
| 22 | study     | 22    | e_coli_316407                            | sweep/codon_cai_bench.py (via<br>sugarcode-ai)            |
| 23 | study     | 23    | bdata.20130222.mhci.txt                  | sweep/neohunter_iedb_bench.py                             |
| 24 | study     | 24    | LP_PDBBind.csv                           | sweep/docking_bench.py                                    |
| 25 | pdb_entry |       | 1CNW                                     | sweep/docking_bench.py                                    |
| 26 | pdb_entry |       | 1EJN                                     | sweep/docking_bench.py                                    |
| 27 | pdb_entry |       | 1F5L                                     | sweep/docking_bench.py                                    |
| 28 | pdb_entry |       | 1G54                                     | sweep/docking_bench.py                                    |
| 29 | pdb_entry |       | 1GI1                                     | sweep/docking_bench.py                                    |
| 30 | pdb_entry |       | 1K1J                                     | sweep/docking_bench.py                                    |
| 31 | pdb_entry |       | 1O3L                                     | sweep/docking_bench.py                                    |
| 32 | pdb_entry |       | 1P1O                                     | sweep/docking_bench.py                                    |
| 33 | pdb_entry |       | 1Y3X                                     | sweep/docking_bench.py                                    |
| 34 | pdb_entry |       | 1ZGI                                     | sweep/docking_bench.py                                    |
| 35 | pdb_entry |       | 2ERZ                                     | sweep/docking_bench.py                                    |
| 36 | pdb_entry |       | 2J4I                                     | sweep/docking_bench.py                                    |
| 37 | pdb_entry |       | 2P4J                                     | sweep/docking_bench.py                                    |
| 38 | pdb_entry |       | 2X7T                                     | sweep/docking_bench.py                                    |
| 39 | pdb_entry |       | 2XAB                                     | sweep/docking_bench.py                                    |
| 40 | pdb_entry |       | 2XJG                                     | sweep/docking_bench.py                                    |
| 41 | pdb_entry |       | 2ZDN                                     | sweep/docking_bench.py                                    |
| 42 | pdb_entry |       | 3B3C                                     | sweep/docking_bench.py                                    |
| 43 | pdb_entry |       | 3C84                                     | sweep/docking_bench.py                                    |
| 44 | pdb_entry |       | 3D8W                                     | sweep/docking_bench.py                                    |
| 45 | pdb_entry |       | 3HKQ                                     | sweep/docking_bench.py                                    |
| 46 | pdb_entry |       | 3HLL                                     | sweep/docking_bench.py                                    |
| 47 | pdb_entry |       | 3IBI                                     | sweep/docking_bench.py                                    |
| 48 | pdb_entry |       | 3M35                                     | sweep/docking_bench.py                                    |
| 49 | pdb_entry |       | 3PO6                                     | sweep/docking_bench.py                                    |
| 50 | pdb_entry |       | 3R17                                     | sweep/docking_bench.py                                    |
| 51 | pdb_entry |       | 3R4N                                     | sweep/docking_bench.py                                    |
| 52 | pdb_entry |       | 4B2I                                     | sweep/docking_bench.py                                    |
| 53 | pdb_entry |       | 4DB7                                     | sweep/docking_bench.py                                    |

| #   | Kind      | Study | Accession                              | Used in                         |
|-----|-----------|-------|----------------------------------------|---------------------------------|
| 54  | pdb_entry |       | 4F5Y                                   | sweep/docking_bench.py          |
| 55  | pdb_entry |       | 4FL1                                   | sweep/docking_bench.py          |
| 56  | pdb_entry |       | 4J93                                   | sweep/docking_bench.py          |
| 57  | pdb_entry |       | 4JFM                                   | sweep/docking_bench.py          |
| 58  | pdb_entry |       | 4KNJ                                   | sweep/docking_bench.py          |
| 59  | pdb_entry |       | 4LXZ                                   | sweep/docking_bench.py          |
| 60  | pdb_entry |       | 4MO8                                   | sweep/docking_bench.py          |
| 61  | pdb_entry |       | 4NH7                                   | sweep/docking_bench.py          |
| 62  | pdb_entry |       | 4O04                                   | sweep/docking_bench.py          |
| 63  | pdb_entry |       | 4OAG                                   | sweep/docking_bench.py          |
| 64  | pdb_entry |       | 4R76                                   | sweep/docking_bench.py          |
| 65  | pdb_entry |       | 4RFC                                   | sweep/docking_bench.py          |
| 66  | pdb_entry |       | 4TIM                                   | sweep/docking_bench.py          |
| 67  | pdb_entry |       | 4XIT                                   | sweep/docking_bench.py          |
| 68  | pdb_entry |       | 4YTC                                   | sweep/docking_bench.py          |
| 69  | pdb_entry |       | 4YXI                                   | sweep/docking_bench.py          |
| 70  | pdb_entry |       | 4ZME                                   | sweep/docking_bench.py          |
| 71  | pdb_entry |       | 5AML                                   | sweep/docking_bench.py          |
| 72  | pdb_entry |       | 5CAP                                   | sweep/docking_bench.py          |
| 73  | pdb_entry |       | 5CAU                                   | sweep/docking_bench.py          |
| 74  | pdb_entry |       | 5D3L                                   | sweep/docking_bench.py          |
| 75  | pdb_entry |       | 5EWK                                   | sweep/docking_bench.py          |
| 76  | pdb_entry |       | 5EWY                                   | sweep/docking_bench.py          |
| 77  | pdb_entry |       | 5F60                                   | sweep/docking_bench.py          |
| 78  | pdb_entry |       | 5F62                                   | sweep/docking_bench.py          |
| 79  | pdb_entry |       | 5G5V                                   | sweep/docking_bench.py          |
| 80  | pdb_entry |       | 5HTZ                                   | sweep/docking_bench.py          |
| 81  | pdb_entry |       | 5IX0                                   | sweep/docking_bench.py          |
| 82  | pdb_entry |       | 5L8C                                   | sweep/docking_bench.py          |
| 83  | pdb_entry |       | 5MJN                                   | sweep/docking_bench.py          |
| 84  | pdb_entry |       | 5MPN                                   | sweep/docking_bench.py          |
| 85  | pdb_entry |       | 5N24                                   | sweep/docking_bench.py          |
| 86  | pdb_entry |       | 5O87                                   | sweep/docking_bench.py          |
| 87  | pdb_entry |       | 5SZ4                                   | sweep/docking_bench.py          |
| 88  | pdb_entry |       | 5SZ5                                   | sweep/docking_bench.py          |
| 89  | pdb_entry |       | 5Y8Y                                   | sweep/docking_bench.py          |
| 90  | pdb_entry |       | 6EQ5                                   | sweep/docking_bench.py          |
| 91  | pdb_entry |       | 6G38                                   | sweep/docking_bench.py          |
| 92  | pdb_entry |       | 6G6T                                   | sweep/docking_bench.py          |
| 93  | pdb_entry |       | 6HQY                                   | sweep/docking_bench.py          |
| 94  | pdb_entry |       | 6HRQ                                   | sweep/docking_bench.py          |
| 95  | pdb_entry |       | 6OE0                                   | sweep/docking_bench.py          |
| 96  | pdb_entry |       | 6RFN                                   | sweep/docking_bench.py          |
| 97  | pdb_entry |       | 6U6W                                   | sweep/docking_bench.py          |
| 98  | species   |       | 10090-ASTROCYTE-integrated.txt         | sweep/codon_cai_multispecies.py |
| 99  | species   |       | GCF.000001635.27                       | sweep/codon_cai_multispecies.py |
| 100 | species   |       | 10116-CARDIAC_VENTRICLE-integrated.txt | sweep/codon_cai_multispecies.py |
| 101 | species   |       | GCF.036323735.1                        | sweep/codon_cai_multispecies.py |
| 102 | species   |       | 121845-WHOLE_ORGANISM-integrated.txt   | sweep/codon_cai_multispecies.py |
| 103 | species   |       | GCF.000475195.1                        | sweep/codon_cai_multispecies.py |
| 104 | species   |       | 1295-WHOLE_ORGANISM-integrated.txt     | sweep/codon_cai_multispecies.py |
| 105 | species   |       | GCF.025558325.1                        | sweep/codon_cai_multispecies.py |
| 106 | species   |       | 1314-WHOLE_ORGANISM-integrated.txt     | sweep/codon_cai_multispecies.py |
| 107 | species   |       | GCF.900475035.1                        | sweep/codon_cai_multispecies.py |
| 108 | species   |       | 185431-WHOLE_ORGANISM-integrated.txt   | sweep/codon_cai_multispecies.py |
| 109 | species   |       | GCF.000002445.2                        | sweep/codon_cai_multispecies.py |

| #   | Kind    | Study | Accession                                    | Used in                         |
|-----|---------|-------|----------------------------------------------|---------------------------------|
| 110 | species |       | 192222-<br>WHOLE_ORGANISM-<br>integrated.txt | sweep/codon_cai_multispecies.py |
| 111 | species |       | GCF_000009085.1                              | sweep/codon_cai_multispecies.py |
| 112 | species |       | 196627-<br>WHOLE_ORGANISM-<br>integrated.txt | sweep/codon_cai_multispecies.py |
| 113 | species |       | GCF_000011325.1                              | sweep/codon_cai_multispecies.py |
| 114 | species |       | 198214-<br>WHOLE_ORGANISM-<br>integrated.txt | sweep/codon_cai_multispecies.py |
| 115 | species |       | GCF_000006925.2                              | sweep/codon_cai_multispecies.py |
| 116 | species |       | 208964-<br>WHOLE_ORGANISM-<br>integrated.txt | sweep/codon_cai_multispecies.py |
| 117 | species |       | GCF_000006765.1                              | sweep/codon_cai_multispecies.py |
| 118 | species |       | 224308-<br>WHOLE_ORGANISM-<br>integrated.txt | sweep/codon_cai_multispecies.py |
| 119 | species |       | GCF_000009045.1                              | sweep/codon_cai_multispecies.py |
| 120 | species |       | 224911-<br>WHOLE_ORGANISM-<br>integrated.txt | sweep/codon_cai_multispecies.py |
| 121 | species |       | GCF_001642675.1                              | sweep/codon_cai_multispecies.py |
| 122 | species |       | 243159-<br>WHOLE_ORGANISM-<br>integrated.txt | sweep/codon_cai_multispecies.py |
| 123 | species |       | GCF_000021485.1                              | sweep/codon_cai_multispecies.py |
| 124 | species |       | 243230-<br>WHOLE_ORGANISM-<br>integrated.txt | sweep/codon_cai_multispecies.py |
| 125 | species |       | GCF_020546685.1                              | sweep/codon_cai_multispecies.py |
| 126 | species |       | 246200-<br>WHOLE_ORGANISM-<br>integrated.txt | sweep/codon_cai_multispecies.py |
| 127 | species |       | GCF_000011965.2                              | sweep/codon_cai_multispecies.py |
| 128 | species |       | 2903-<br>WHOLE_ORGANISM-<br>integrated.txt   | sweep/codon_cai_multispecies.py |
| 129 | species |       | GCF_000372725.1                              | sweep/codon_cai_multispecies.py |
| 130 | species |       | 343509-<br>WHOLE_ORGANISM-<br>integrated.txt | sweep/codon_cai_multispecies.py |
| 131 | species |       | GCF_000010085.1                              | sweep/codon_cai_multispecies.py |
| 132 | species |       | 36329-<br>WHOLE_ORGANISM-<br>integrated.txt  | sweep/codon_cai_multispecies.py |
| 133 | species |       | GCF_000002765.6                              | sweep/codon_cai_multispecies.py |
| 134 | species |       | 403677-<br>WHOLE_ORGANISM-<br>integrated.txt | sweep/codon_cai_multispecies.py |
| 135 | species |       | GCF_000142945.1                              | sweep/codon_cai_multispecies.py |
| 136 | species |       | 4081-PERICARP-<br>integrated.txt             | sweep/codon_cai_multispecies.py |
| 137 | species |       | GCF_036512215.1                              | sweep/codon_cai_multispecies.py |
| 138 | species |       | 4097-LEAF-integrated.txt                     | sweep/codon_cai_multispecies.py |
| 139 | species |       | GCF_000715075.1                              | sweep/codon_cai_multispecies.py |
| 140 | species |       | 418985-<br>WHOLE_ORGANISM-<br>integrated.txt | sweep/codon_cai_multispecies.py |
| 141 | species |       | GCA_002081605.1                              | sweep/codon_cai_multispecies.py |
| 142 | species |       | 44689-<br>WHOLE_ORGANISM-<br>integrated.txt  | sweep/codon_cai_multispecies.py |
| 143 | species |       | GCF_000004695.1                              | sweep/codon_cai_multispecies.py |

| #   | Kind              | Study   | Accession                                               | Used in                         |
|-----|-------------------|---------|---------------------------------------------------------|---------------------------------|
| 144 | species           |         | 4932-<br>WHOLE_ORGANISM-<br>integrated.txt              | sweep/codon_cai_multispecies.py |
| 145 | species           |         | GCF_000146045.2                                         | sweep/codon_cai_multispecies.py |
| 146 | species           |         | 54291-<br>WHOLE_ORGANISM-<br>integrated.txt             | sweep/codon_cai_multispecies.py |
| 147 | species           |         | GCF_030505655.1                                         | sweep/codon_cai_multispecies.py |
| 148 | species           |         | 6239-<br>WHOLE_ORGANISM-<br>integrated.txt              | sweep/codon_cai_multispecies.py |
| 149 | species           |         | GCF_000002985.6                                         | sweep/codon_cai_multispecies.py |
| 150 | species           |         | 7227-SPERM-integrated.txt                               | sweep/codon_cai_multispecies.py |
| 151 | species           |         | GCF_000001215.4                                         | sweep/codon_cai_multispecies.py |
| 152 | species           |         | 7460-ABDOMEN-<br>integrated.txt                         | sweep/codon_cai_multispecies.py |
| 153 | species           |         | GCF_003254395.2                                         | sweep/codon_cai_multispecies.py |
| 154 | species           |         | 8030-LIVER-integrated.txt                               | sweep/codon_cai_multispecies.py |
| 155 | species           |         | GCF_965601325.2                                         | sweep/codon_cai_multispecies.py |
| 156 | species           |         | 83332-<br>WHOLE_ORGANISM-<br>integrated.txt             | sweep/codon_cai_multispecies.py |
| 157 | species           |         | GCF_000195955.2                                         | sweep/codon_cai_multispecies.py |
| 158 | species           |         | 9031-HEART-integrated.txt                               | sweep/codon_cai_multispecies.py |
| 159 | species           |         | GCF_016699485.2                                         | sweep/codon_cai_multispecies.py |
| 160 | species           |         | 93061-<br>WHOLE_ORGANISM-<br>integrated.txt             | sweep/codon_cai_multispecies.py |
| 161 | species           |         | GCF_000013425.1                                         | sweep/codon_cai_multispecies.py |
| 162 | species           |         | 9544-BILE_DUCT-<br>integrated.txt                       | sweep/codon_cai_multispecies.py |
| 163 | species           |         | GCF_049350105.2                                         | sweep/codon_cai_multispecies.py |
| 164 | species           |         | 9606-ADRENAL_GLAND-<br>integrated.txt                   | sweep/codon_cai_multispecies.py |
| 165 | species           |         | GCF_000001405.40                                        | sweep/codon_cai_multispecies.py |
| 166 | species           |         | 9615-PLASMA-<br>integrated.txt                          | sweep/codon_cai_multispecies.py |
| 167 | species           |         | GCF_011100685.1                                         | sweep/codon_cai_multispecies.py |
| 168 | species           |         | 9823-LIVER-integrated.txt                               | sweep/codon_cai_multispecies.py |
| 169 | species           |         | GCF_054392235.1                                         | sweep/codon_cai_multispecies.py |
| 170 | species           |         | 9913-<br>LENS_OF_CAMERA_TYPE_EYE-<br>integrated.txt     | sweep/codon_cai_multispecies.py |
| 171 | species           |         | GCF_002263795.3                                         | sweep/codon_cai_multispecies.py |
| 172 | species           |         | 99287-<br>WHOLE_ORGANISM-<br>integrated.txt             | sweep/codon_cai_multispecies.py |
| 173 | species           |         | GCF_000006945.2                                         | sweep/codon_cai_multispecies.py |
| 174 | species           |         | 9986-<br>PERIPHERAL_REGION_OF_RETINA-<br>integrated.txt | sweep/codon_cai_multispecies.py |
| 175 | species           |         | GCF_964237555.1                                         | sweep/codon_cai_multispecies.py |
| 176 | gtrnadb           |         | Esch.coli_K_12.MG1655                                   | sweep/codon_tai_enc.py          |
| 177 | gtrnadb           |         | Baci_subt_subtilis_168                                  | sweep/codon_tai_enc.py          |
| 178 | gtrnadb           |         | Scere3                                                  | sweep/codon_tai_enc.py          |
| 179 | refseq_transcript | 25      | NM_000546.6                                             | sweep/primer_vs_primer3.py      |
| 180 | refseq_transcript | 25      | NM_007294.4                                             | sweep/primer_vs_primer3.py      |
| 181 | refseq_transcript | 25      | NM_000059.4                                             | sweep/primer_vs_primer3.py      |
| 182 | refseq_transcript | 25      | NM_004333.6                                             | sweep/primer_vs_primer3.py      |
| 183 | refseq_transcript | 25      | NM_005228.5                                             | sweep/primer_vs_primer3.py      |
| 184 | study             | 26      | clinvar.vcf.gz                                          | sweep/vcf_vs_pysam.py           |
| 185 | study             | 27      | GSE52778                                                | sweep/de_vs_pydeseq2.py         |
| 186 | rfam_family       | RF00004 | RF00004                                                 | sweep/stockholm_vs_biopython.py |
| 187 | rfam_family       | RF00015 | RF00015                                                 | sweep/stockholm_vs_biopython.py |
| 188 | rfam_family       | RF00017 | RF00017                                                 | sweep/stockholm_vs_biopython.py |
| 189 | rfam_family       | RF00023 | RF00023                                                 | sweep/stockholm_vs_biopython.py |
| 190 | rfam_family       | RF00050 | RF00050                                                 | sweep/stockholm_vs_biopython.py |

| #   | Kind          | Study                | Accession                | Used in                                        |
|-----|---------------|----------------------|--------------------------|------------------------------------------------|
| 191 | rfam_family   | RF00059              | RF00059                  | sweep/stockholm_vs_biopython.py                |
| 192 | rfam_family   | RF00162              | RF00162                  | sweep/stockholm_vs_biopython.py                |
| 193 | rfam_family   | RF00167              | RF00167                  | sweep/stockholm_vs_biopython.py                |
| 194 | rfam_family   | RF00168              | RF00168                  | sweep/stockholm_vs_biopython.py                |
| 195 | rfam_family   | RF00174              | RF00174                  | sweep/stockholm_vs_biopython.py                |
| 196 | rfam_family   | RF00234              | RF00234                  | sweep/stockholm_vs_biopython.py                |
| 197 | rfam_family   | RF00504              | RF00504                  | sweep/stockholm_vs_biopython.py                |
| 198 | refseq_genome | J01749.1             | J01749.1                 | sweep/proteinprops_restriction_vs_biopython.py |
| 199 | refseq_genome | NC_001416.1          | NC_001416.1              | sweep/proteinprops_restriction_vs_biopython.py |
| 200 | opentree_node | ott1062253           | ott1062253               | sweep/newick_vs_biophylo.py                    |
| 201 | opentree_node | ott392223            | ott392223                | sweep/newick_vs_biophylo.py                    |
| 202 | opentree_node | ott44565             | ott44565                 | sweep/newick_vs_biophylo.py                    |
| 203 | opentree_node | ott691846            | ott691846                | sweep/newick_vs_biophylo.py                    |
| 204 | opentree_node | ott81461             | ott81461                 | sweep/newick_vs_biophylo.py                    |
| 205 | opentree_node | ott913935            | ott913935                | sweep/newick_vs_biophylo.py                    |
| 206 | study         | NA12878_chr20_lowcov | NA12878.chrom20.ILLUMINA | sweep/tilt_low_coverage.20121211.bam           |
| 207 | study         | SRR622461            | SRR622461                | sweep/fastq_vs_biopython.py                    |

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